

Bibliometria na Prática com *R* e *Bibliometrix*

Coleta de dados nas bases *Scopus* e *Web of Science*

Clayton Martins Pereira
Doutorando em Ciência da Informação
FFC/UNESP

dezembro/2022

Sobre o instrutor:
Clayton Martins Pereira

- Doutorando em Ciência da Informação na FFC/UNESP;
- Mestrado em Informática pelo Instituto Tecnológico de Aeronáutica - ITA;
- Servidor da Divisão de Biblioteca do Instituto Nacional de Pesquisas Espaciais – INPE;
- Administrador da plataforma de biblioteca digital do INPE;
- Atua na apuração de indicadores e na preservação da produção intelectual do INPE;
- Experiência na elaboração de estudos bibliométricos.

Objetivo do curso

Apoiar a elaboração de revisões sistemáticas de literatura por meio da aplicação de:

- técnicas de coleta e tratamento de dados bibliográficos das bases *Scopus* e *Web of Science* (WoS);
- ferramentas para visualização e análise bibliométrica (*Science mapping*) dos dados coletados;



Bibliometria (Science mapping)

Método quantitativo (medição estatística) para mapeamento e avaliação de uma área de pesquisa a partir dos dados bibliográficos de sua produção científica.

A bibliometria possibilita ao pesquisador identificar:

- os artigos mais citados da área;
- os periódicos da área com maior fator de impacto;
- os autores mais citados e mais produtivos;
- os termos e conceitos mais relevantes;
- a produção por período, país e instituição;
- redes de colaboração (autores, países e instituições);
- lacunas/oportunidades de pesquisa na área.

Tópicos a serem abordados

1. Coleta de dados na *Scopus* e na *WoS*

- ✓ Definição dos critérios de busca e das palavras-chaves (Keywords);
- ✓ Elaboração de expressões de busca;
- ✓ Exportação dos resultados de buscas.

2. Tratamento de dados em *R*

- ✓ Elaboração de script no *R Studio* (pacote *Bibliometrix*) para unificação, seleção e sumarização dos dados coletados.

3. Análise Bibliométrica com *Bibliometrix*

- ✓ Visualização de dados na ferramenta *biblioshiny* (pacote *Bibliometrix*).

Programação

1° bloco	Coleta de dados nas bases *WoS* e *Scopus*
	Configuração do ambiente *R Studio*
2° bloco	Tratamento dos dados coletados (*script* em *R*)
	Visualização e análise bibliométrica (*Bibliometrix*)

Exemplo utilizado neste curso

PERGUNTA DE PESQUISA (fictícia):

- De que forma **Métodos e Metodologias Ágeis** estão sendo integrados na pesquisa e desenvolvimento no campo da **Inteligência Artificial**?

CRITÉRIOS DE BUSCA (em inglês):

- Palavras-chave: *dica - consultar tesauros (vocabulários controlados) da área.*
- Bases de busca:
- Tipo de publicação:
- Período de publicação (*se necessário*):

Exemplo utilizado neste curso

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CRITÉRIOS DE BUSCA (em inglês):

- Palavras-chave: "***Agile Method***", "***Agile Methodology***" e "***Artificial Intelligence***";
- Bases de busca: ***Scopus*** e ***Web of Science (WoS)***;
- Tipo de publicação: Artigos em periódicos (***journal article***) e artigos em conferências (***conference article***)
- Período de publicação: últimos 10 anos (**2012 – 2022**)
  Obs: ano corrente + 10 anos completos anteriores!

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Exemplo utilizado neste curso

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CRITÉRIOS DE BUSCA (em inglês):

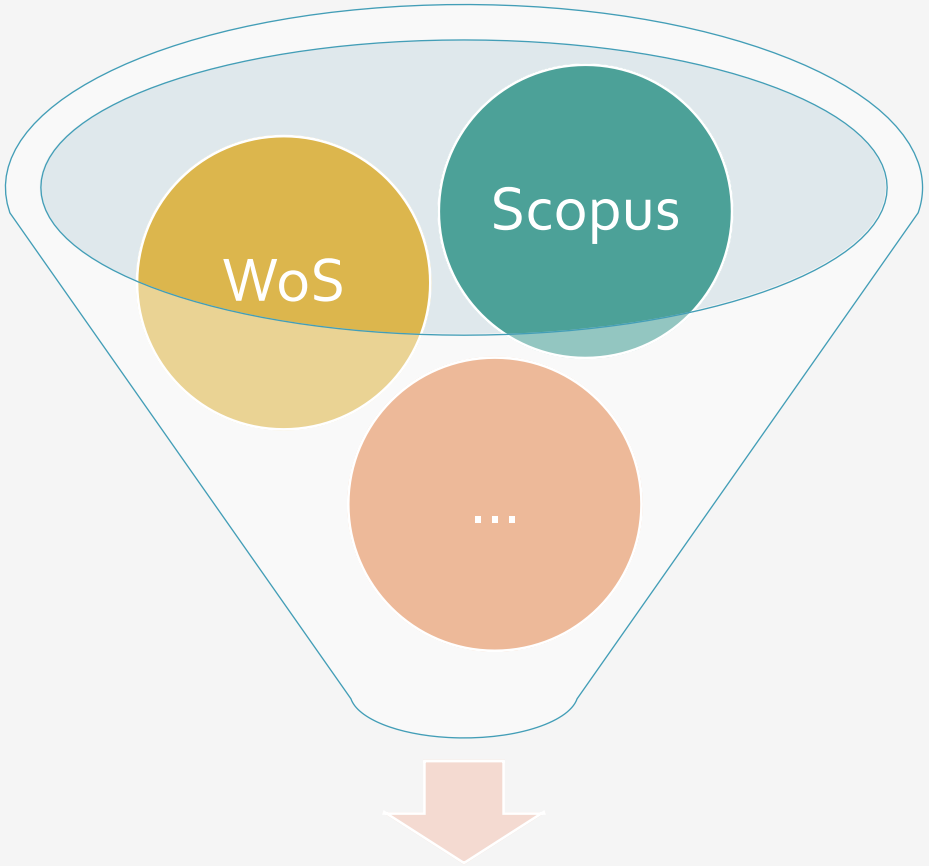
- Palavras-chave: "***Agile Method***", "***Agile Methodology***" e "***Artificial Intelligence***";
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- Tipo de publicação: Artigos em periódicos (***journal article***) e artigos em conferências (***conference article***)
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EXPRESSÃO DE BUSCA:

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✓ **"agile method*" AND "artificial intelligence"**

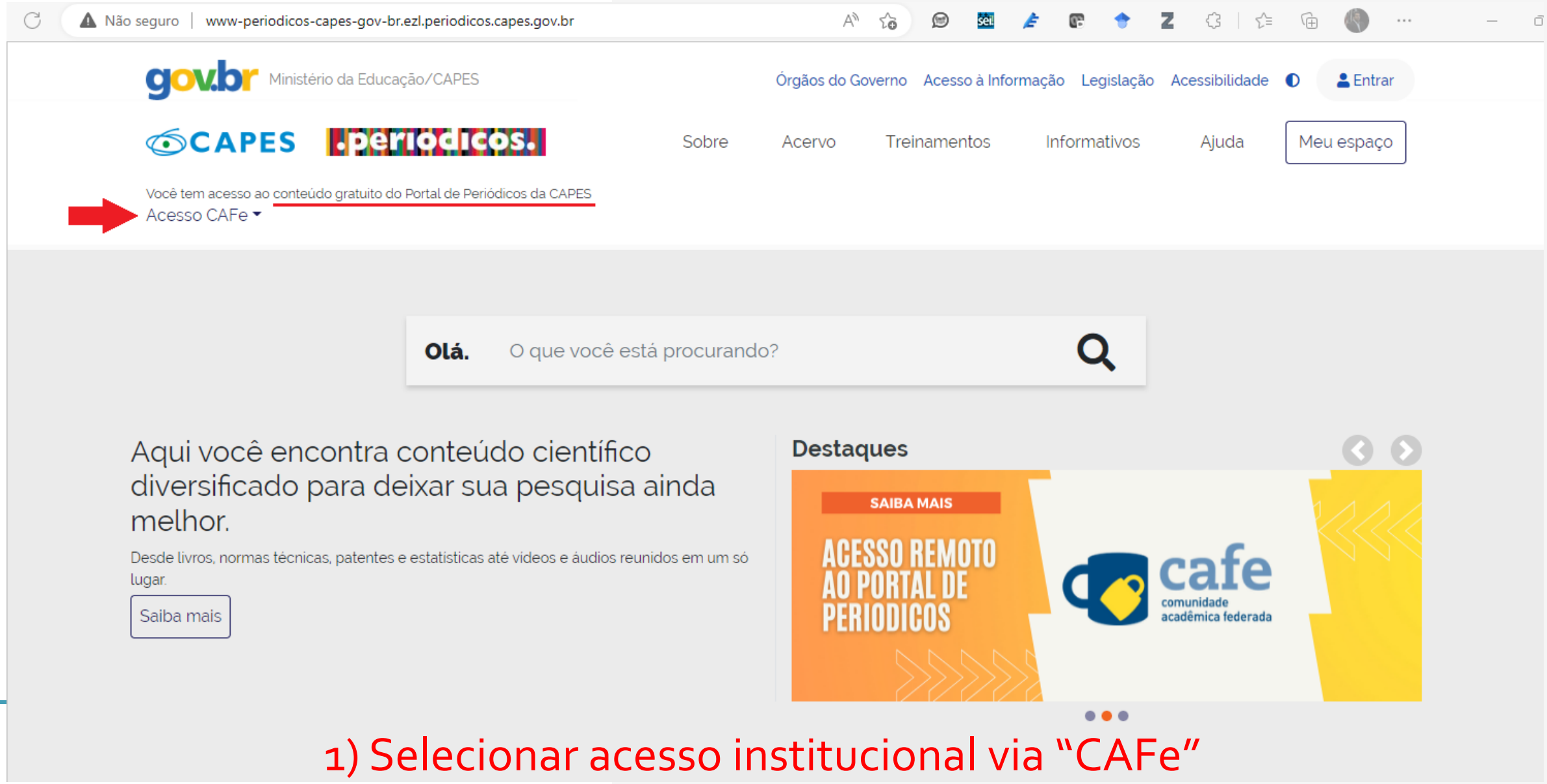
Coleta de dados



Dados Bibliográficos

Acesso ao portal de Periódicos CAPES

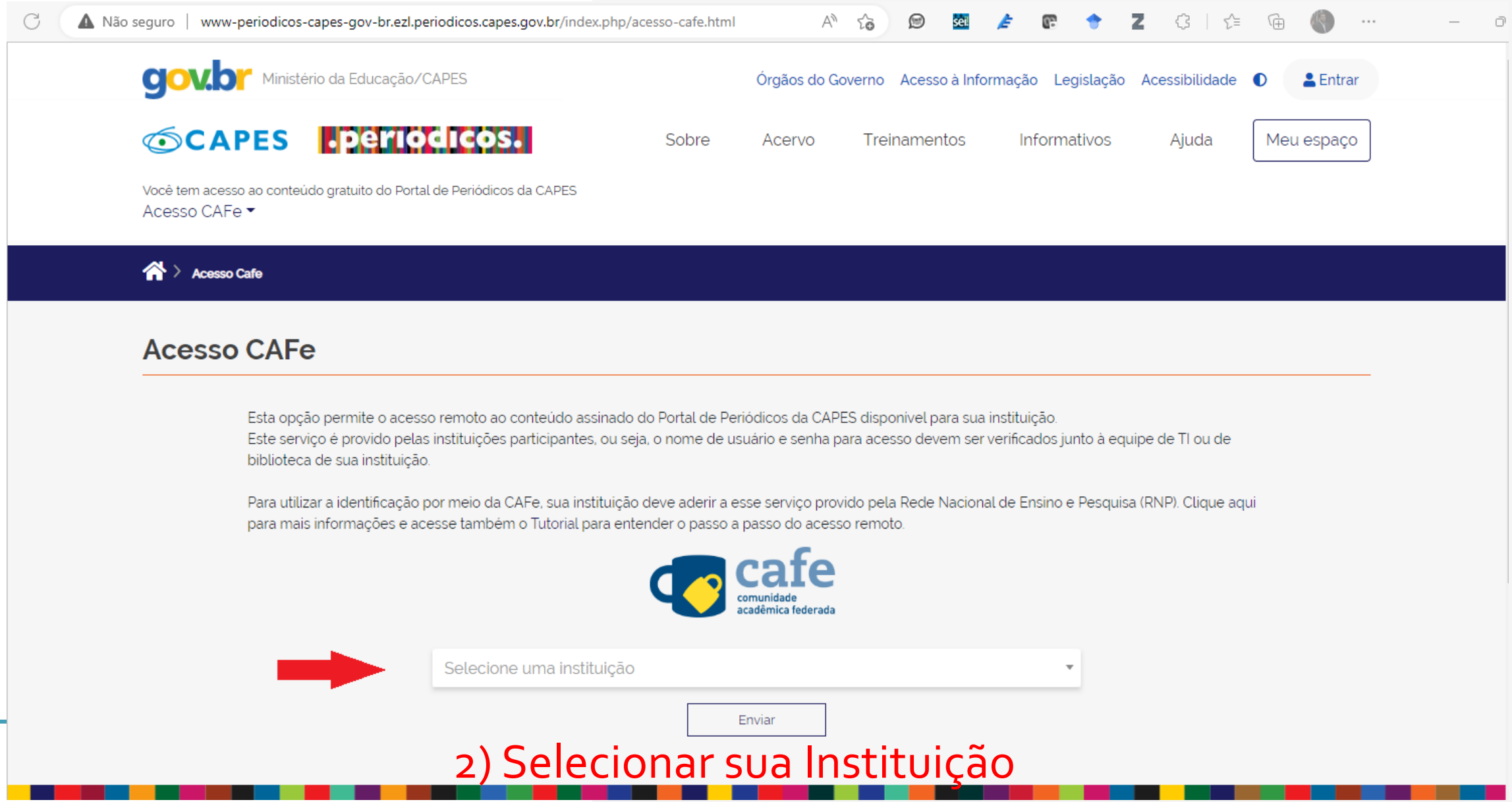
periodicos.capes.gov.br



1) Selecionar acesso institucional via "CAFe"

Acesso ao portal de Periódicos CAPES

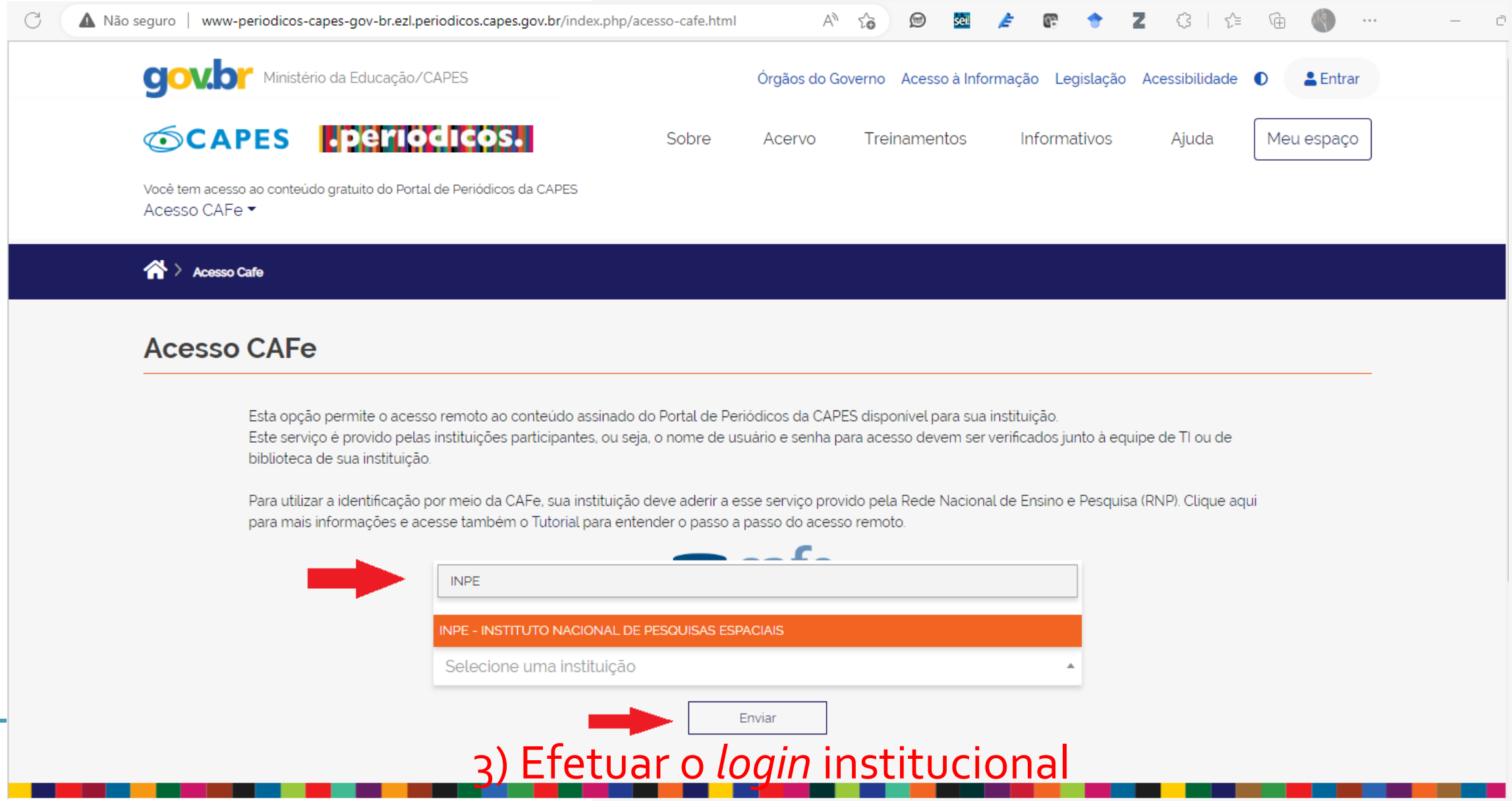
periodicos.capes.gov.br



2) Selecionar sua Instituição

Acesso ao portal de Periódicos CAPES

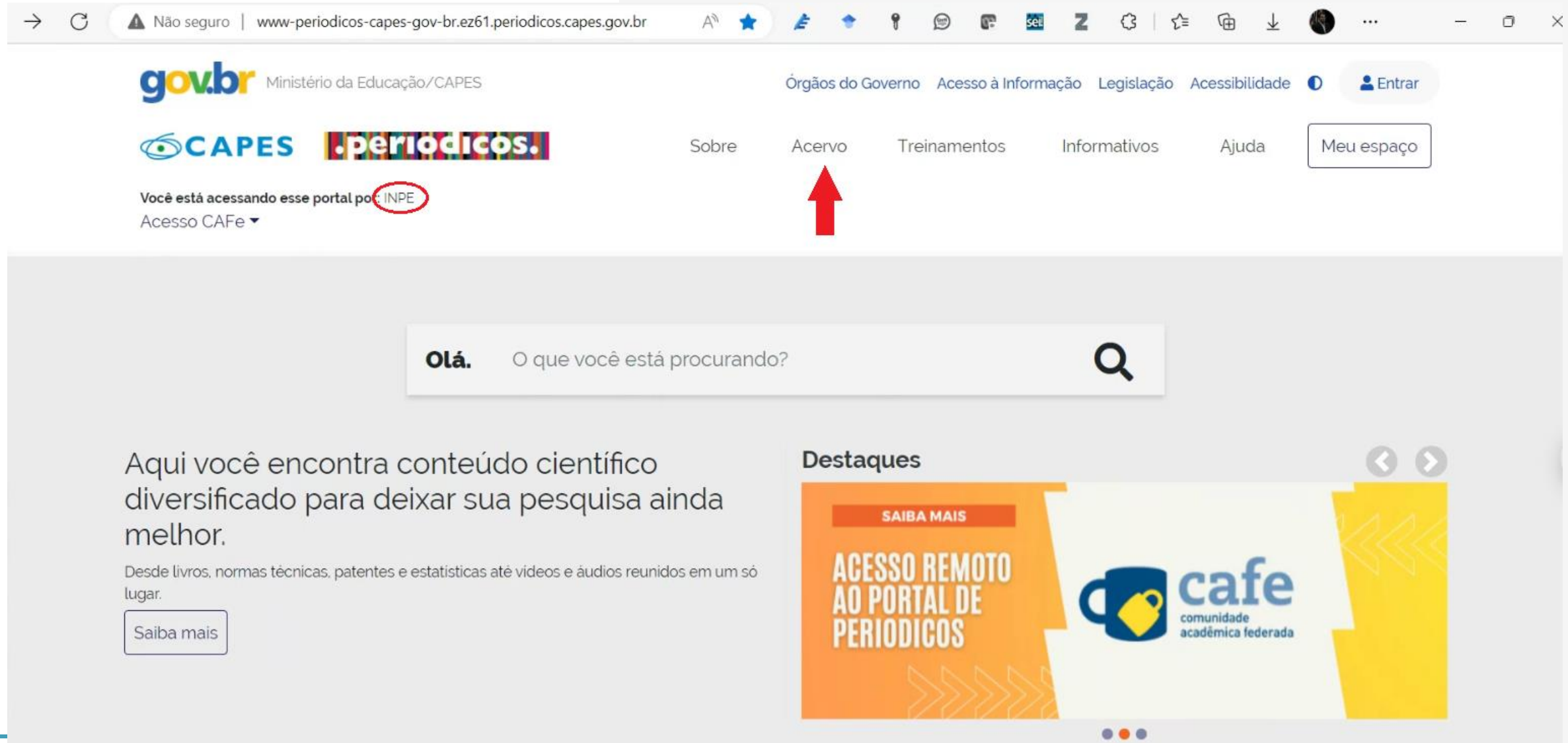
periodicos.capes.gov.br



3) Efetuar o *login* institucional

Acesso ao portal de Periódicos CAPES

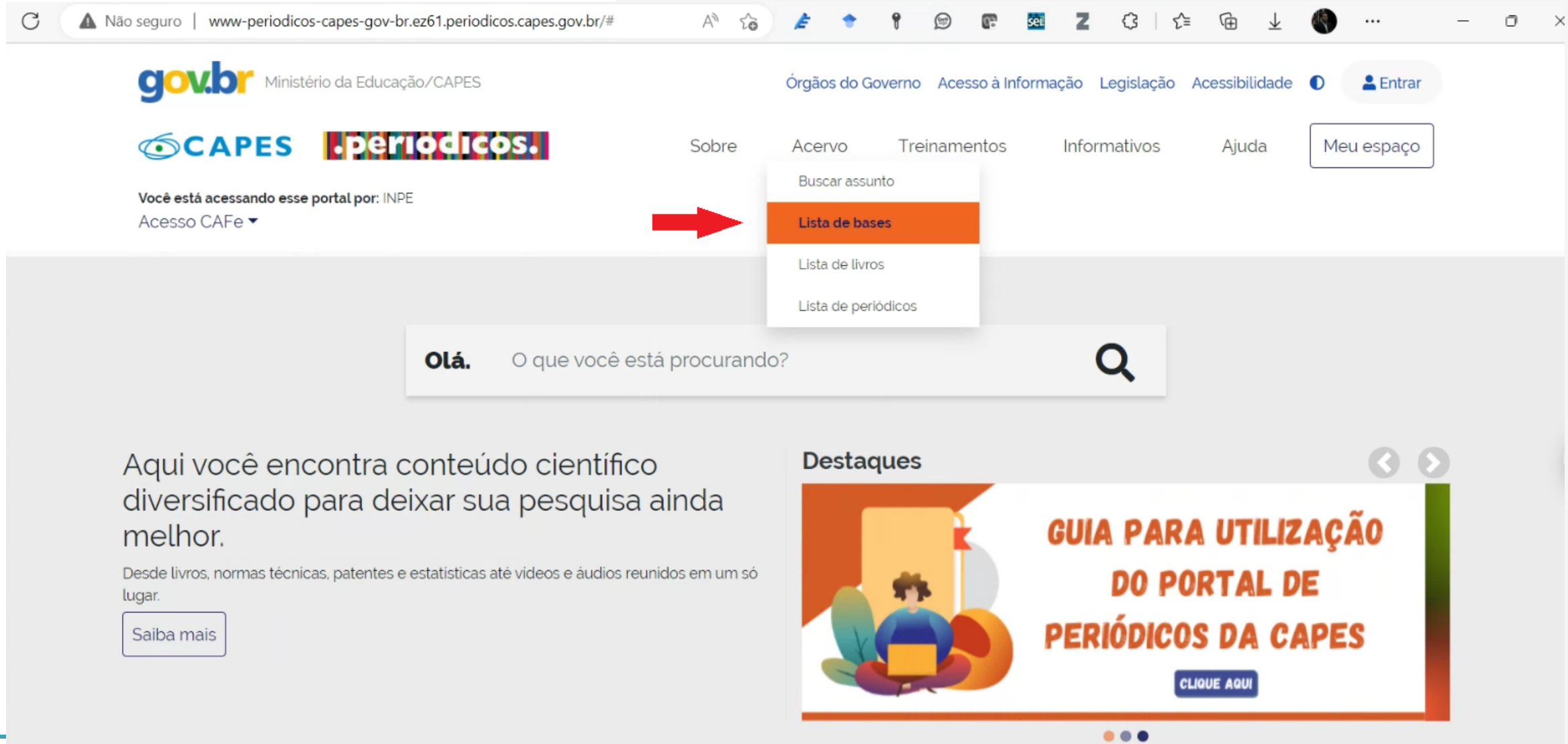
periodicos.capes.gov.br



4) Efetuado o login, clicar na opção "Acervo"

Acesso ao portal de Periódicos CAPES

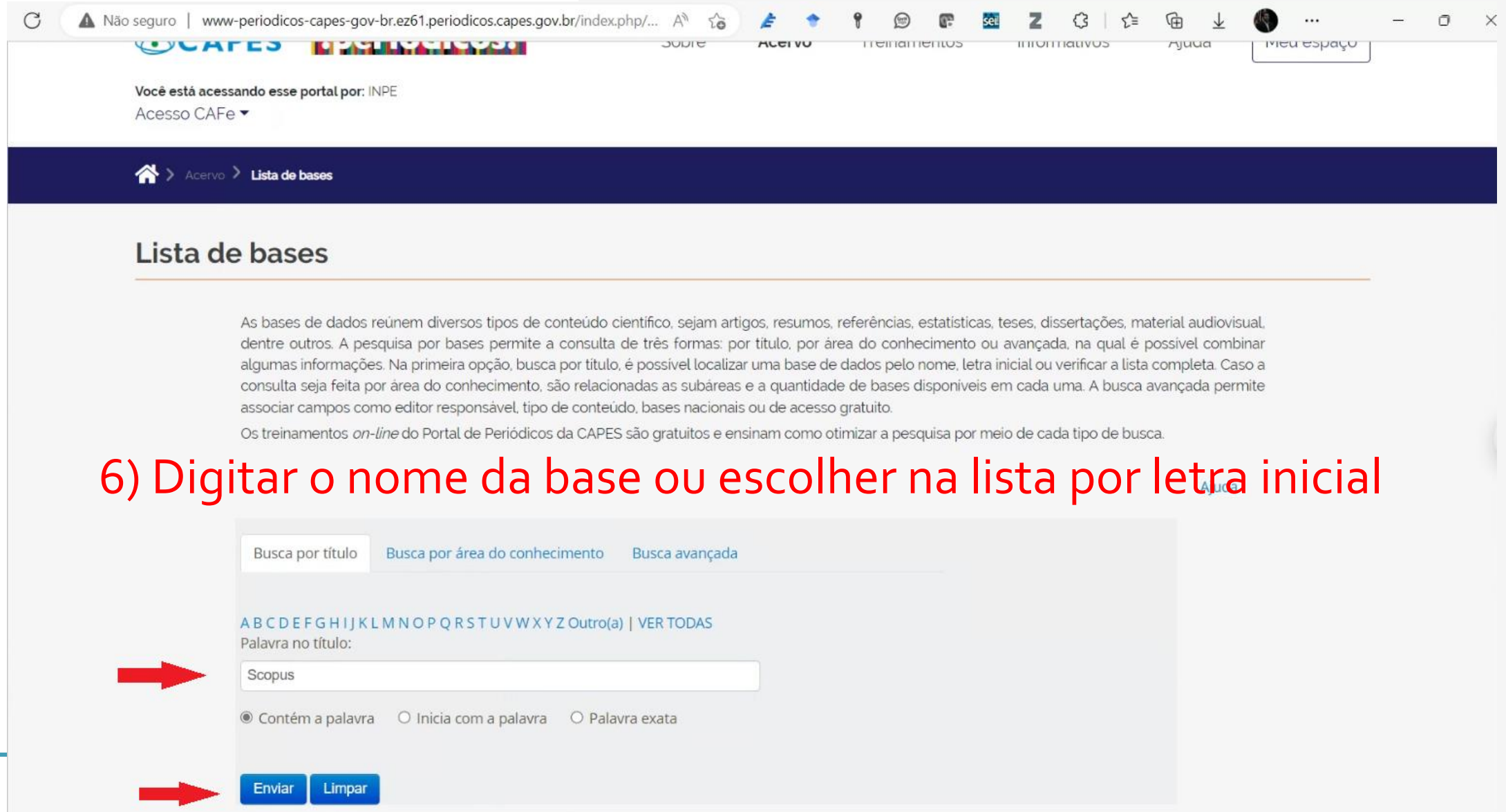
periodicos.capes.gov.br



5) Em "Acervo", escolher a opção "Lista de bases"

Acesso ao portal de Periódicos CAPES

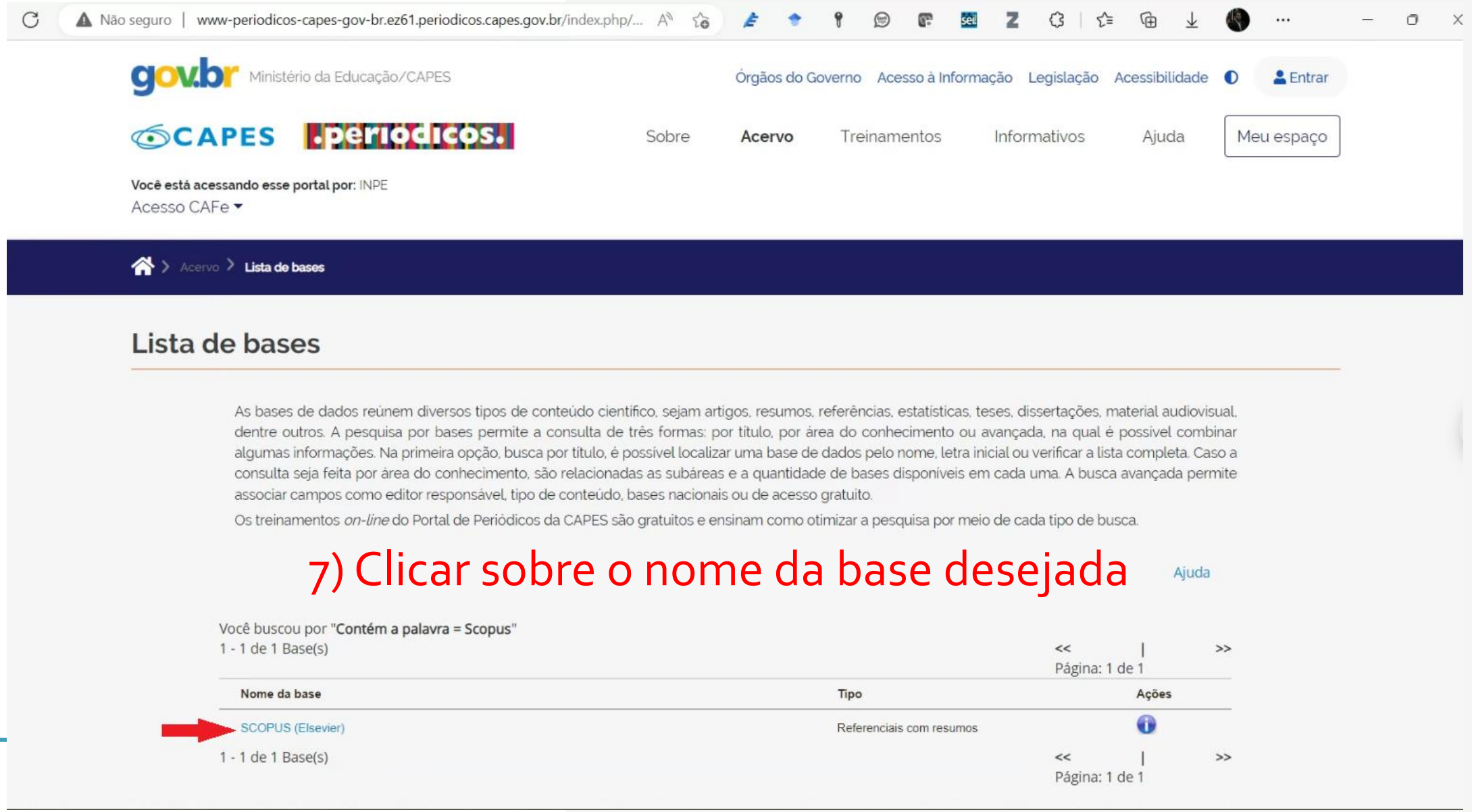
periodicos.capes.gov.br



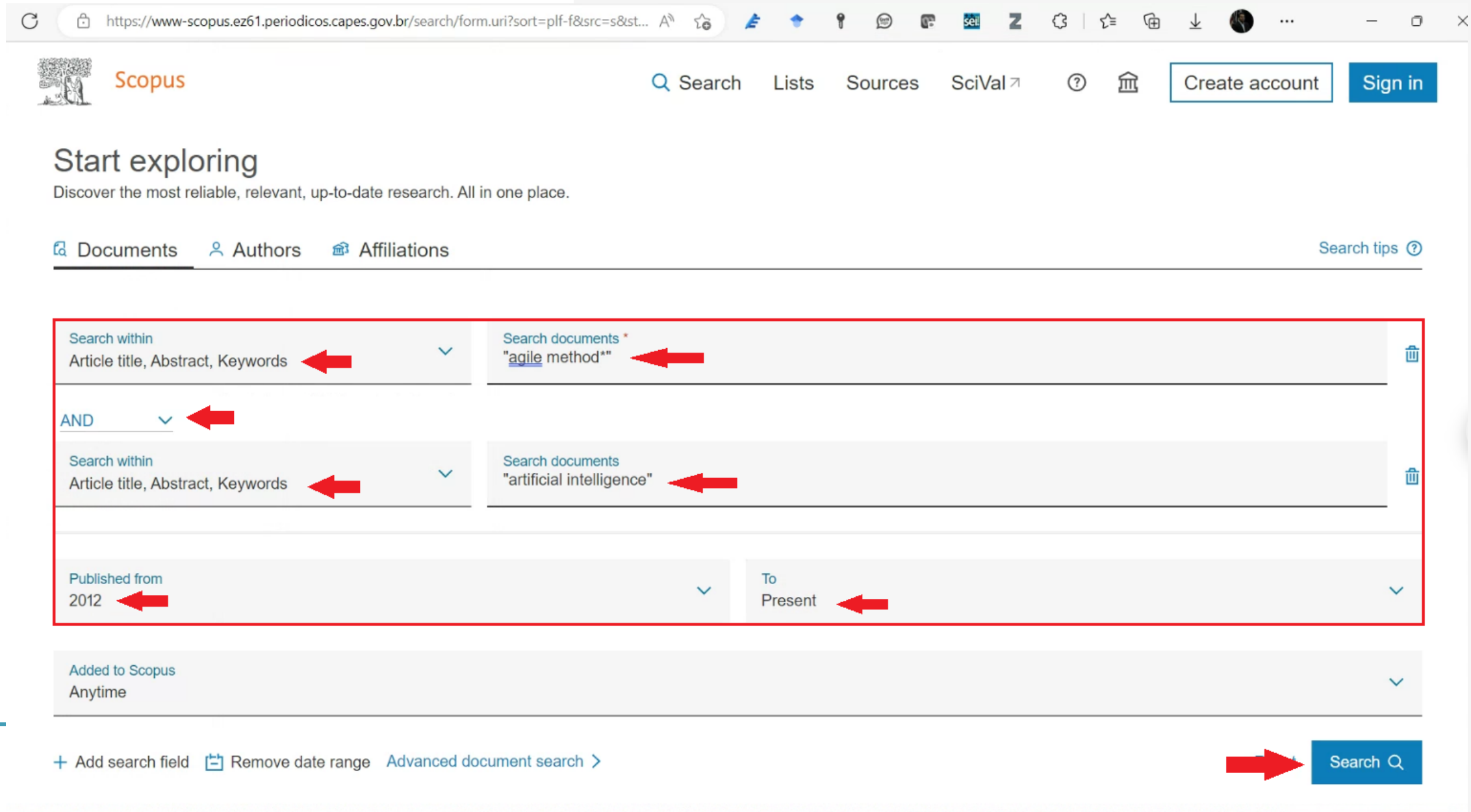
6) Digitar o nome da base ou escolher na lista por letra inicial

Acesso ao portal de Periódicos CAPES

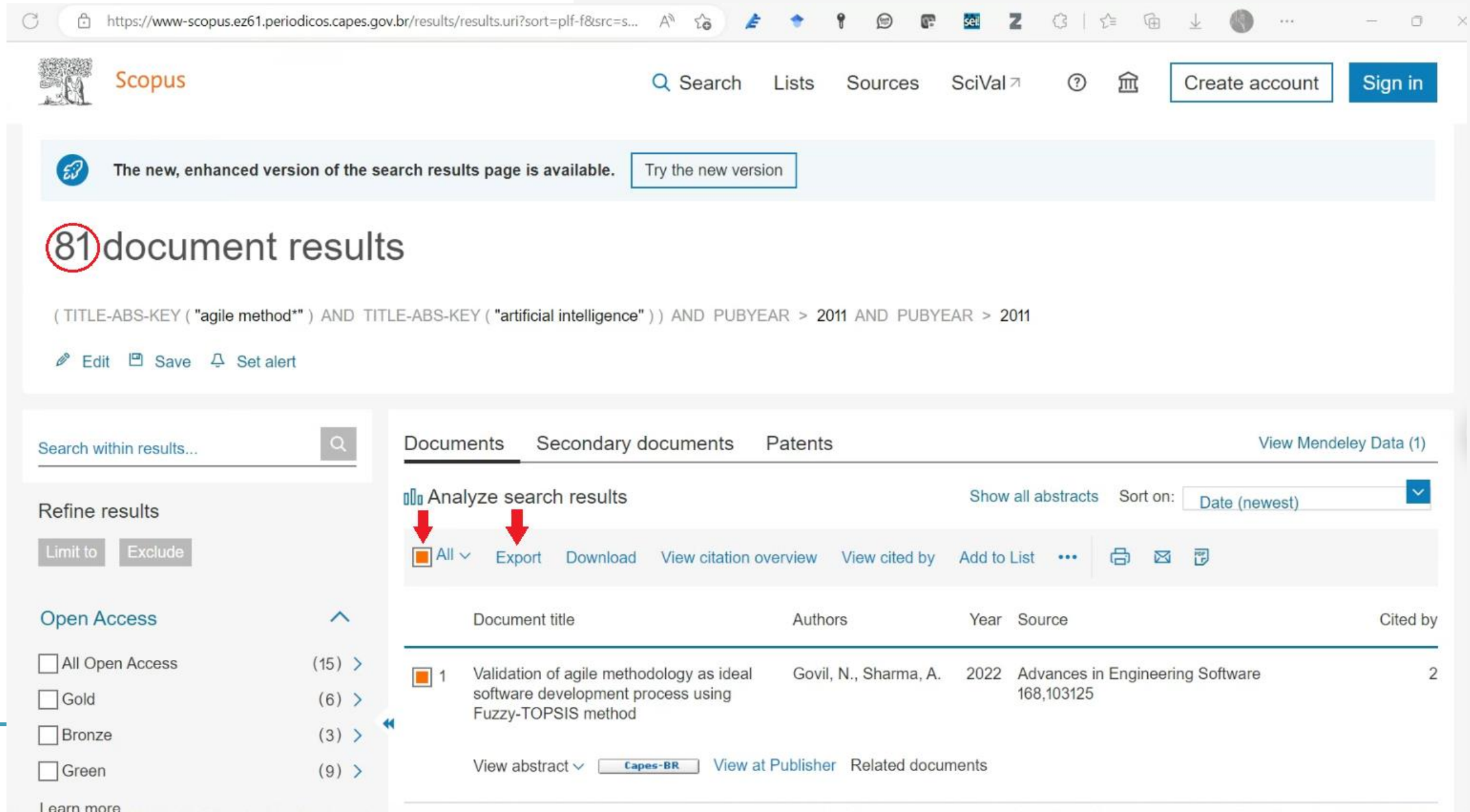
periodicos.capes.gov.br



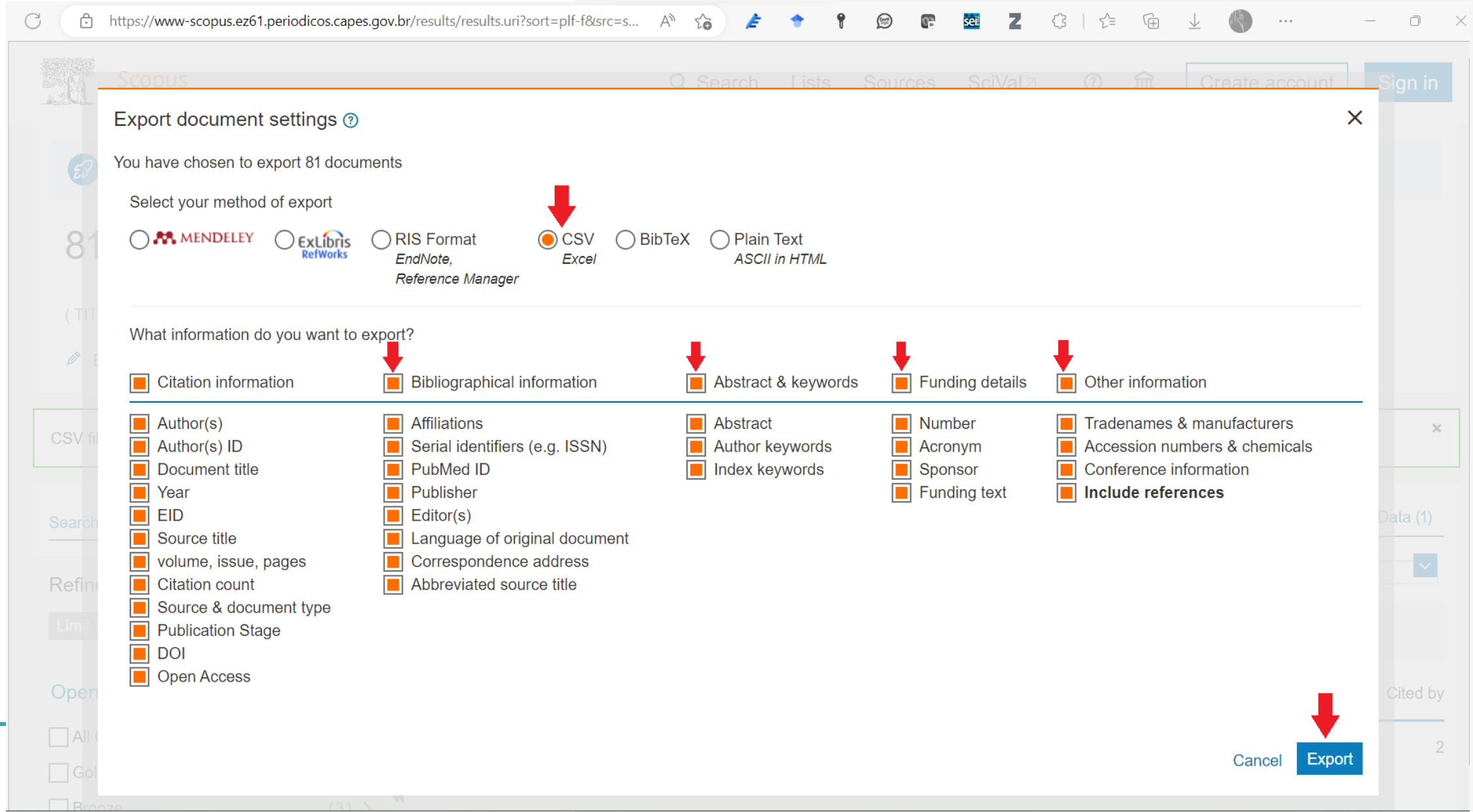
Busca na Scopus



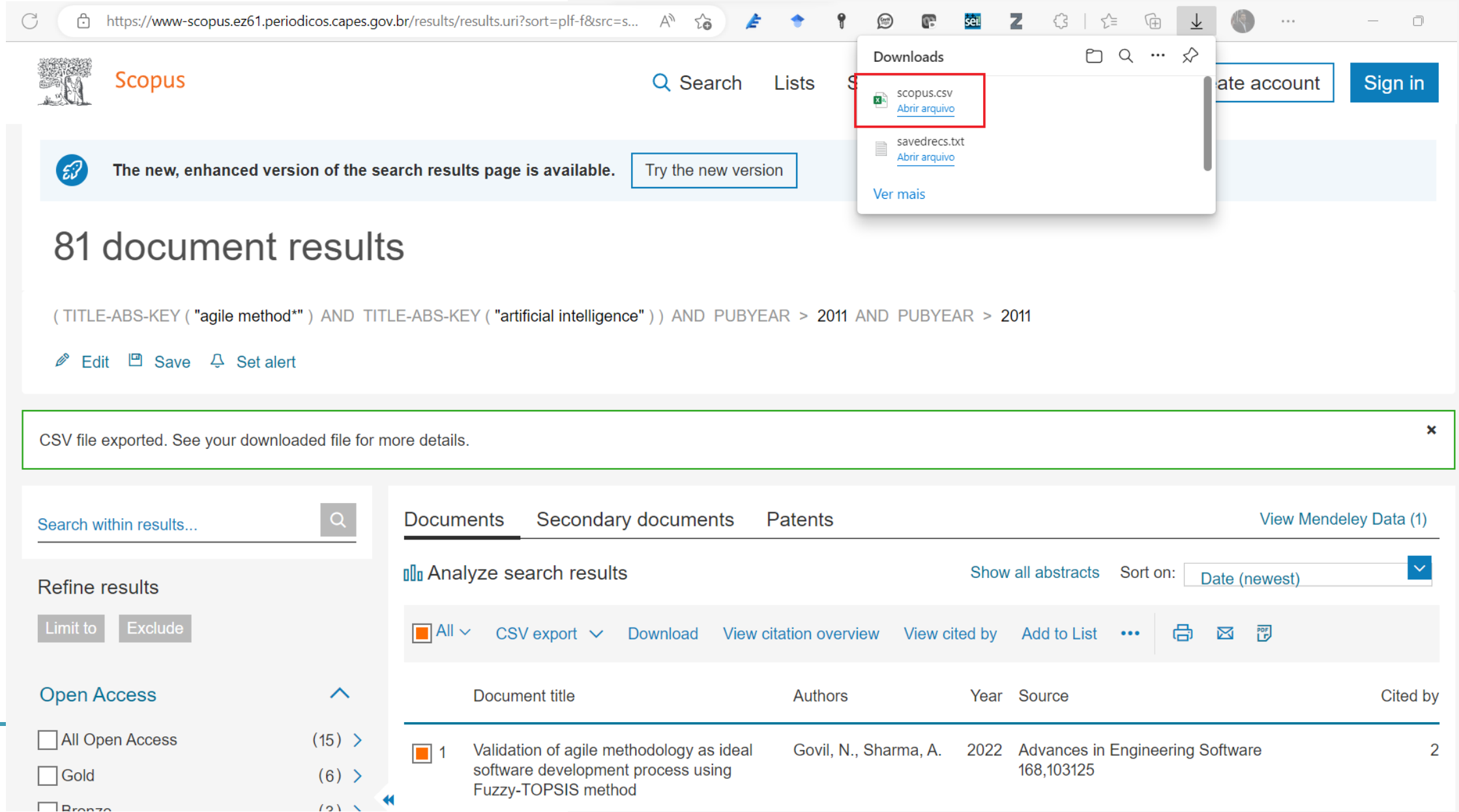
Exportar os resultados da busca



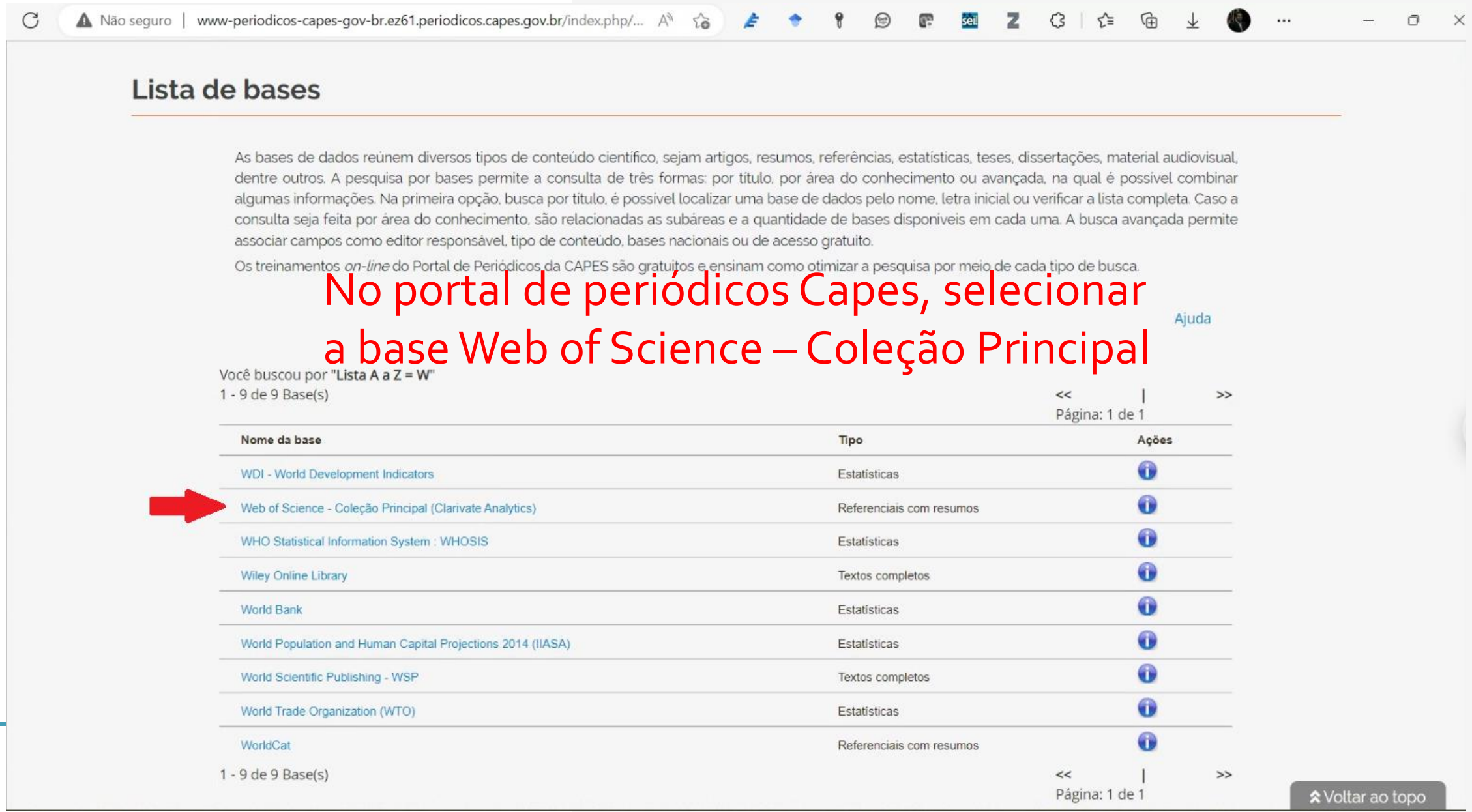
Exportar todos os metadados no formato CSV



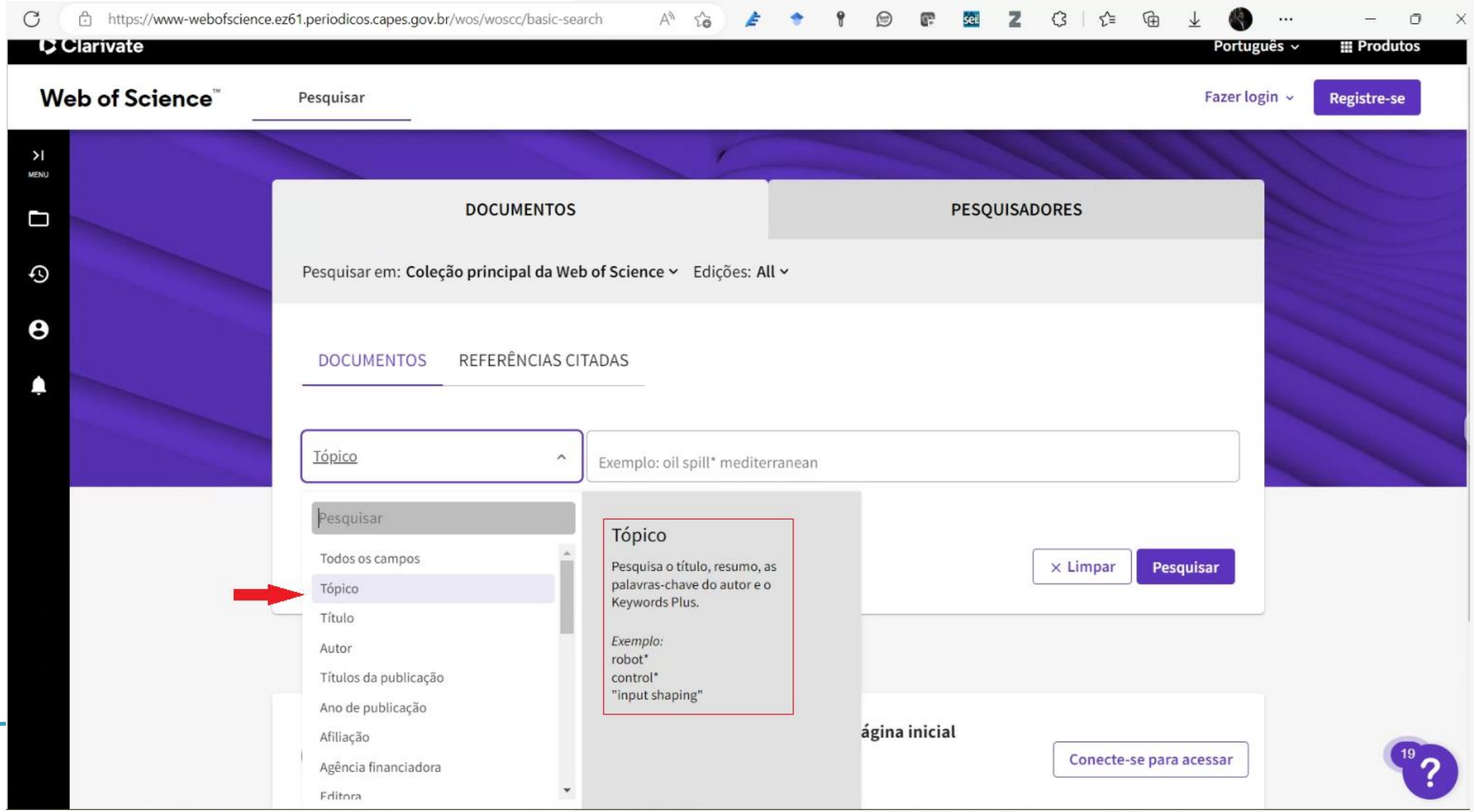
Resultados foram salvos no arquivo scopus.csv



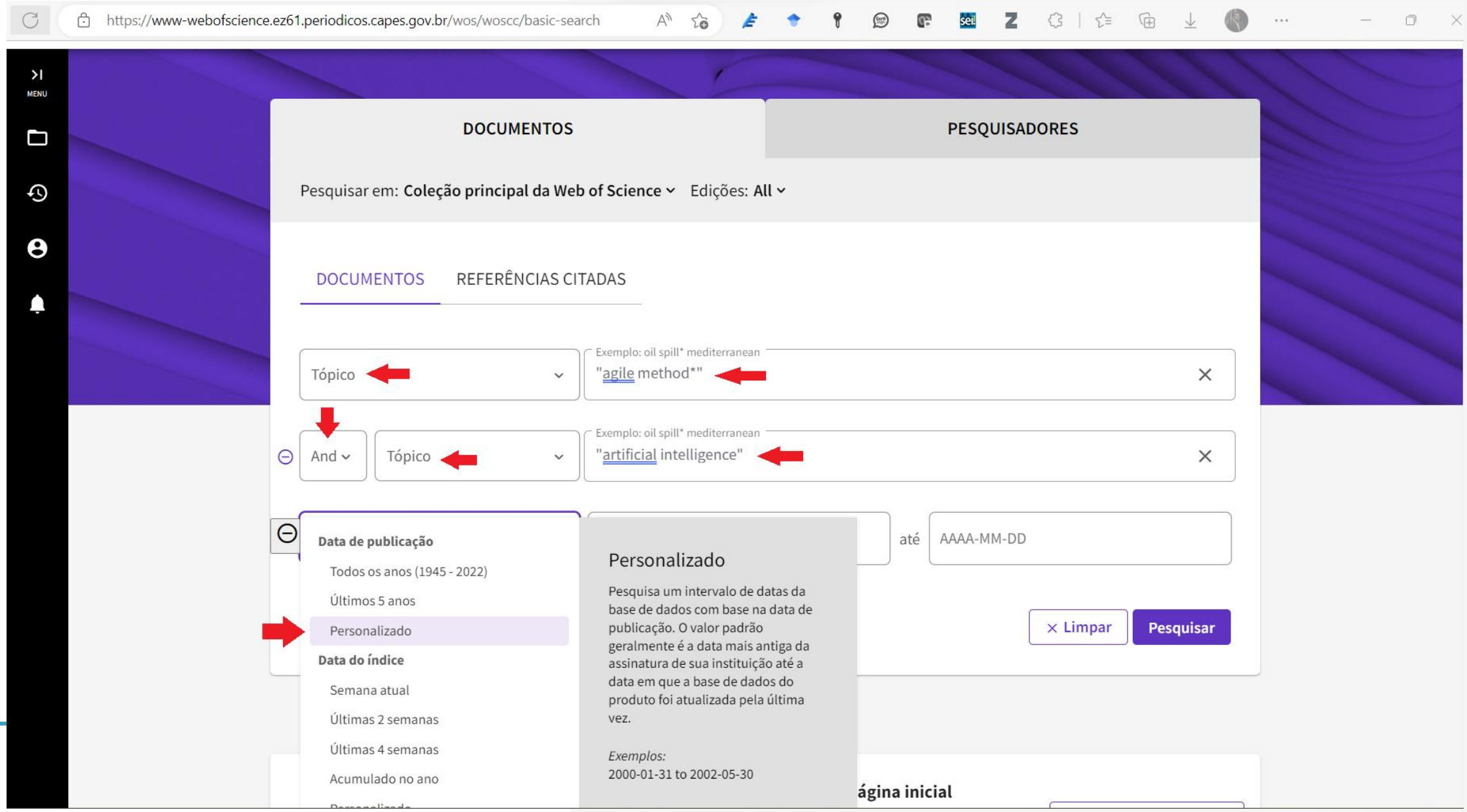
Busca na Web of Science



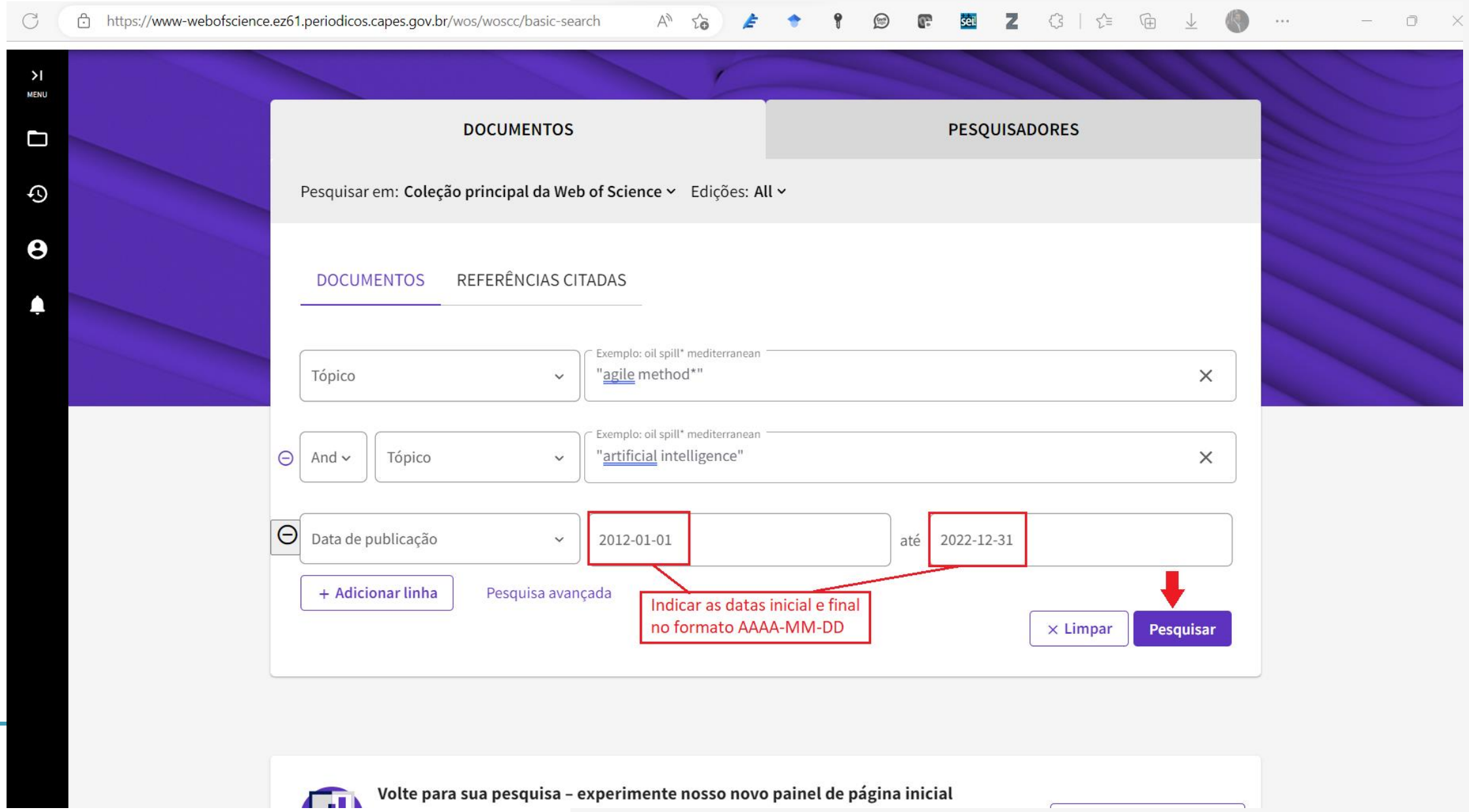
Busca na Web of Science



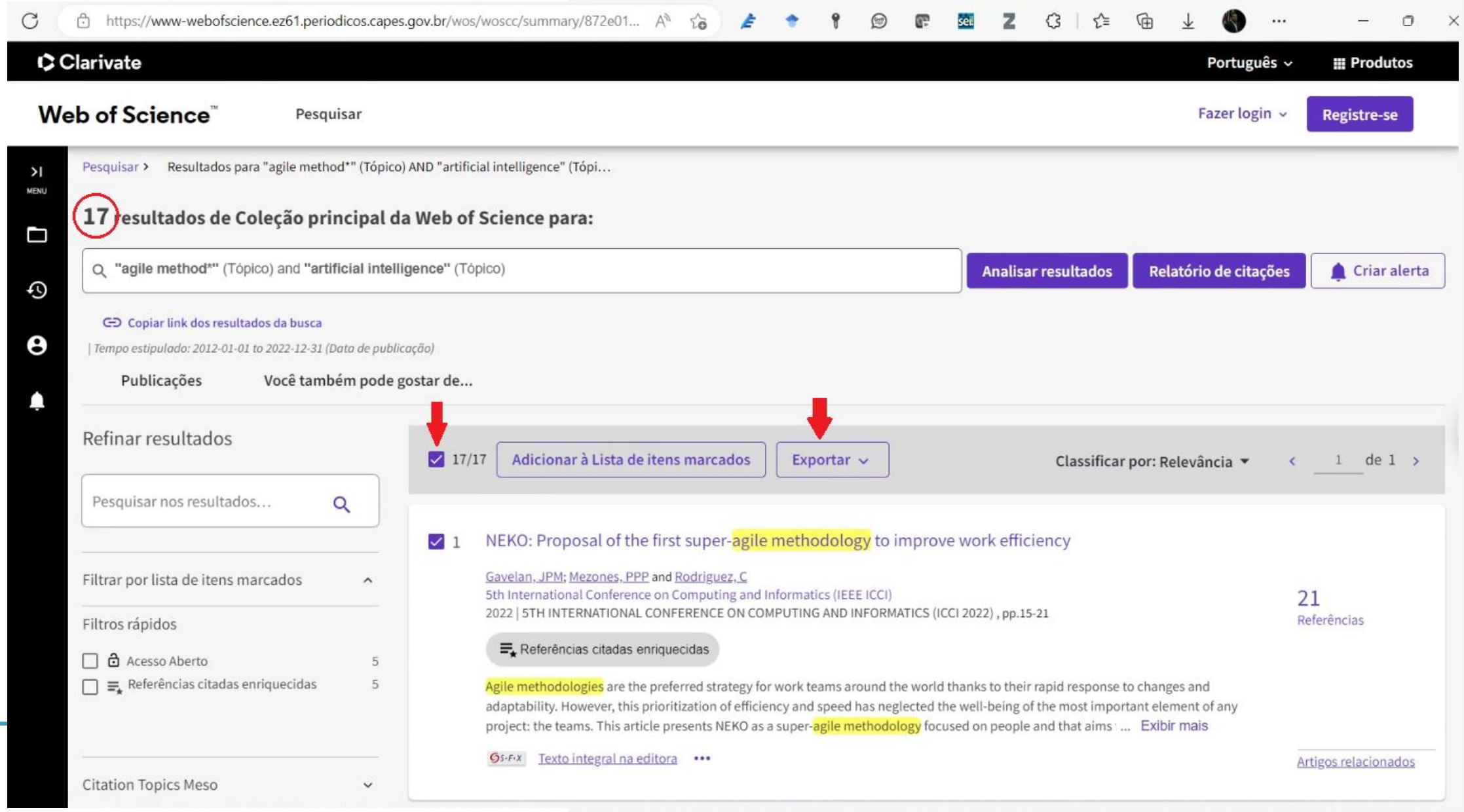
Busca na Web of Science



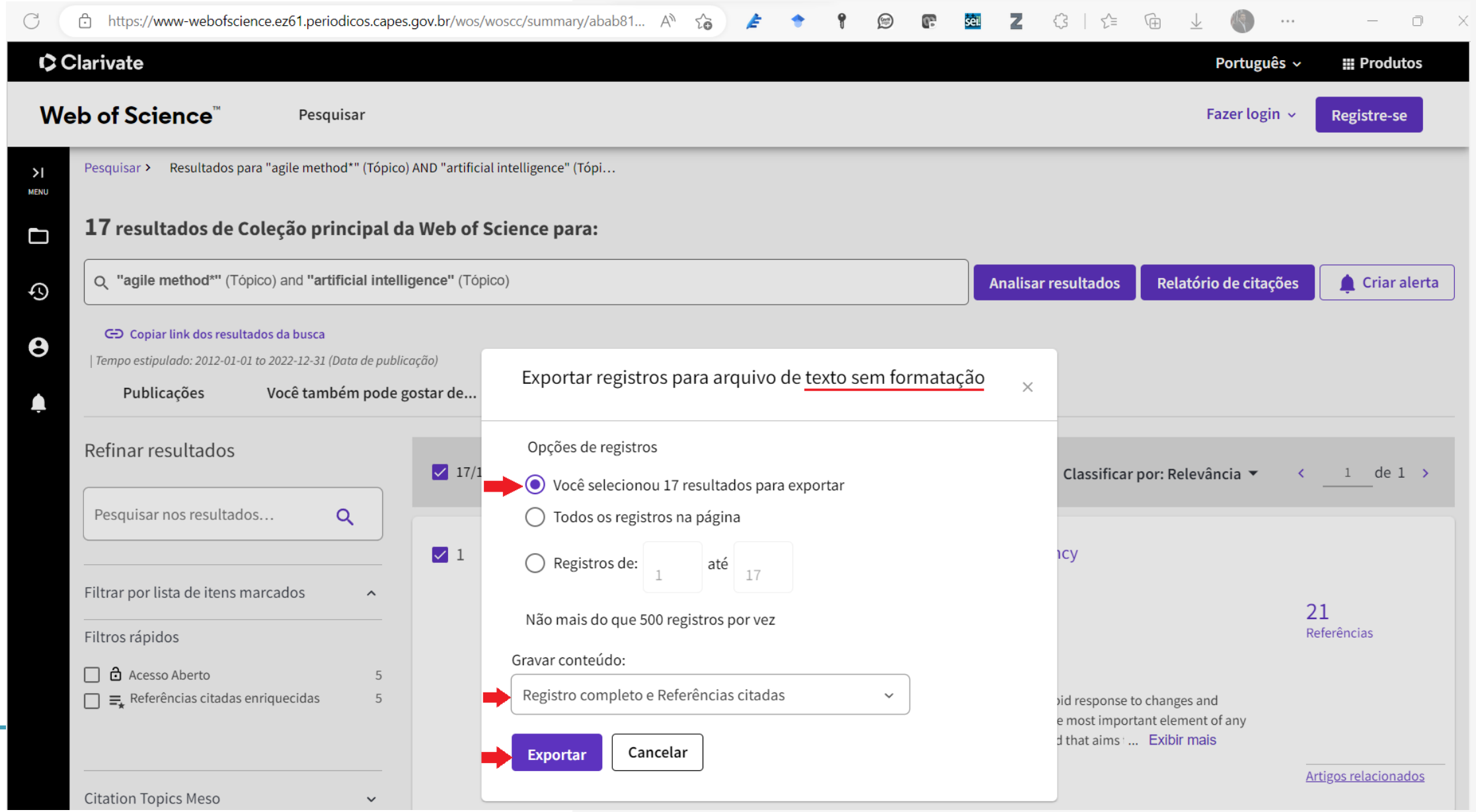
Busca na Web of Science



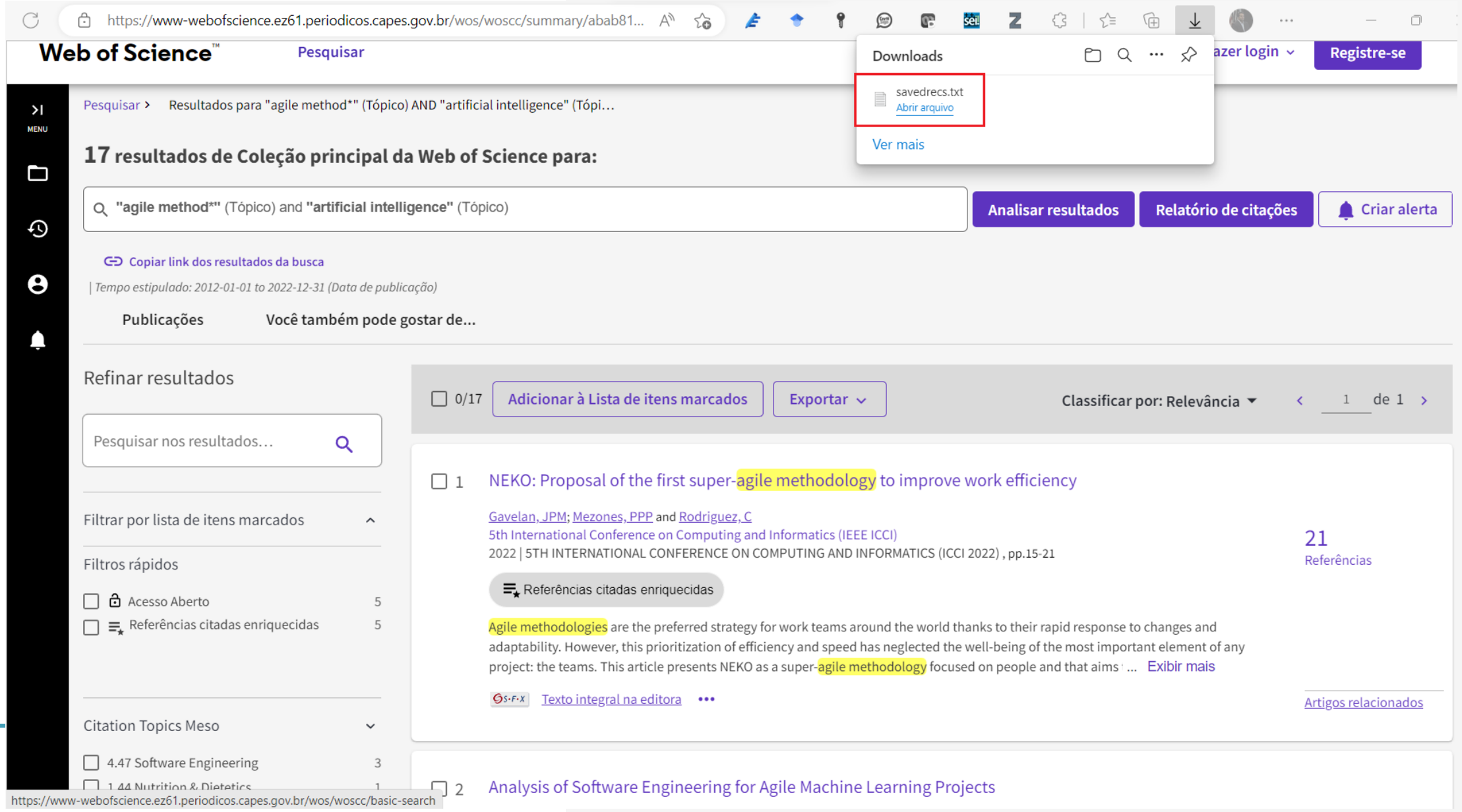
Exportar os resultados da busca no formato "Texto sem formatação"



Exportar o "Registro completo e Referências citadas"

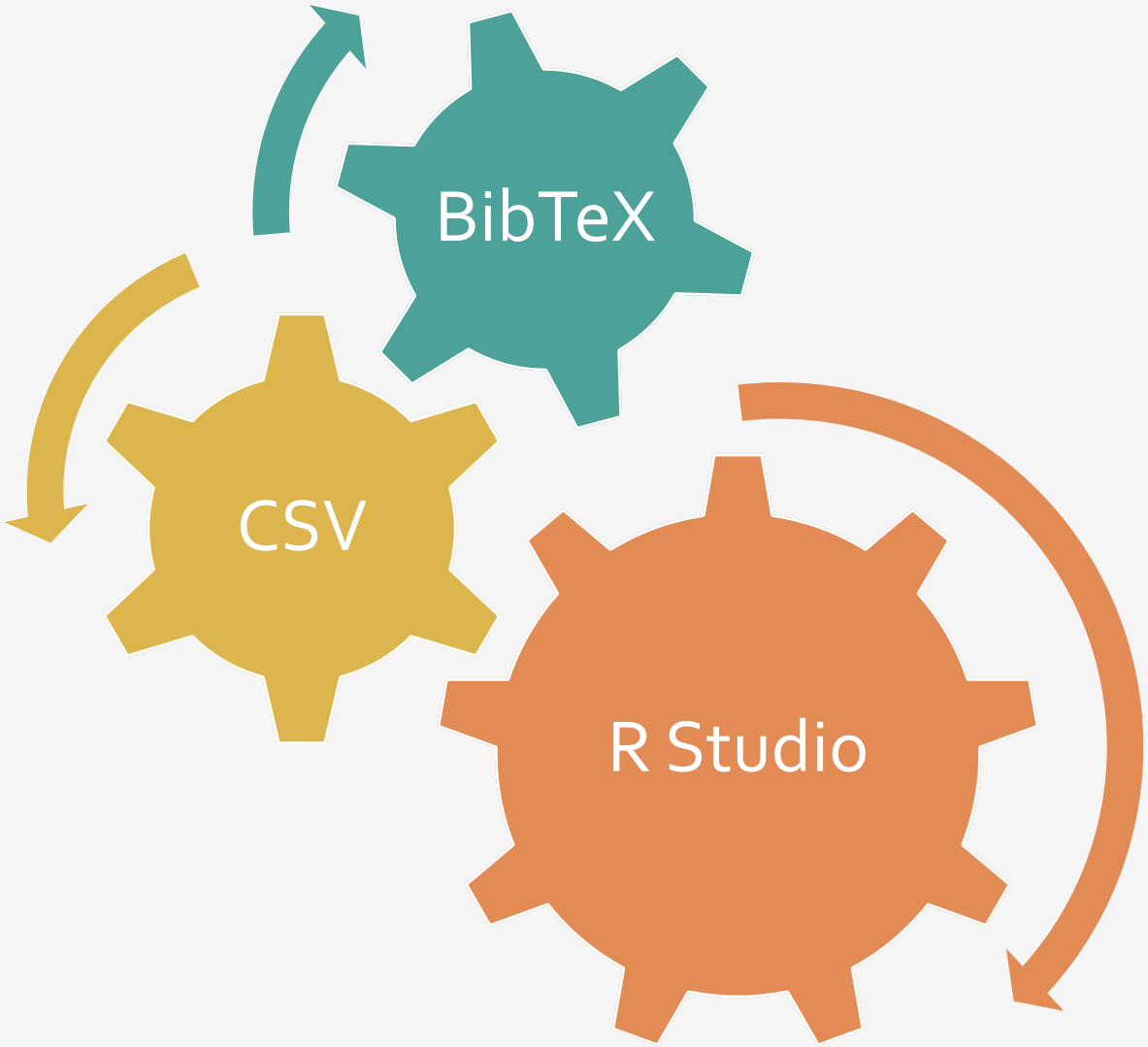


Resultados foram salvos no arquivo savedrecs.txt

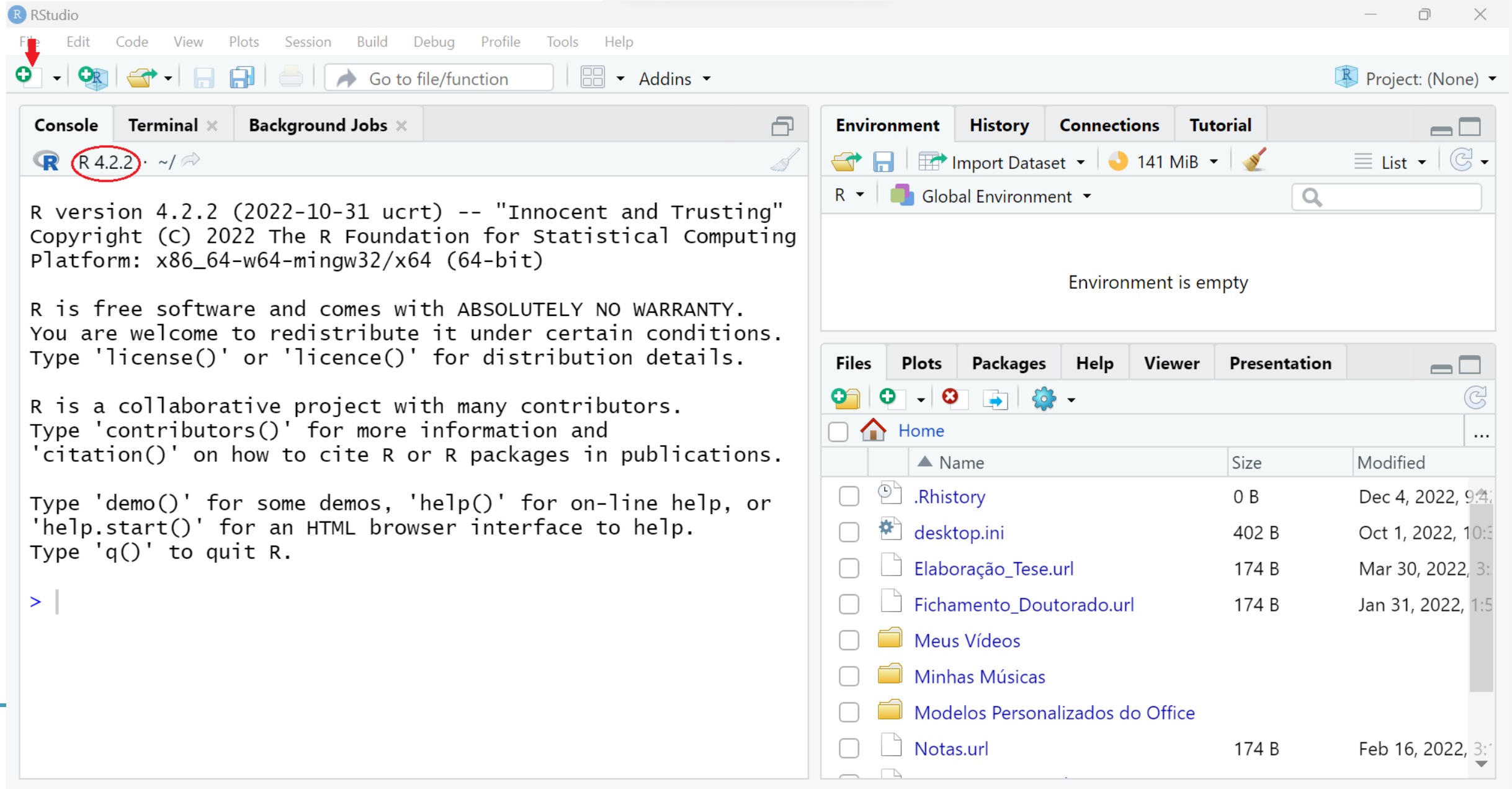


Tratamento dos dados

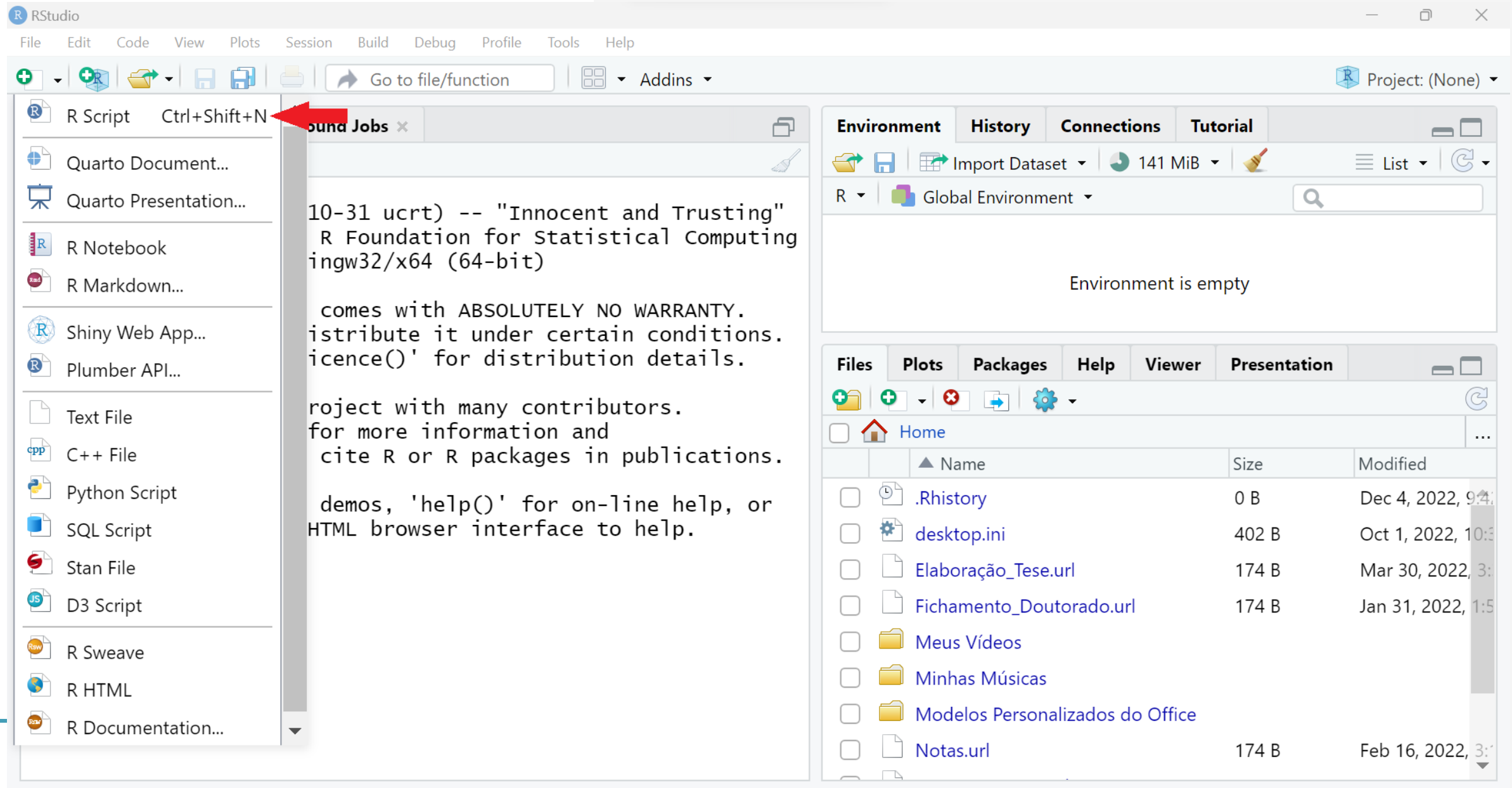




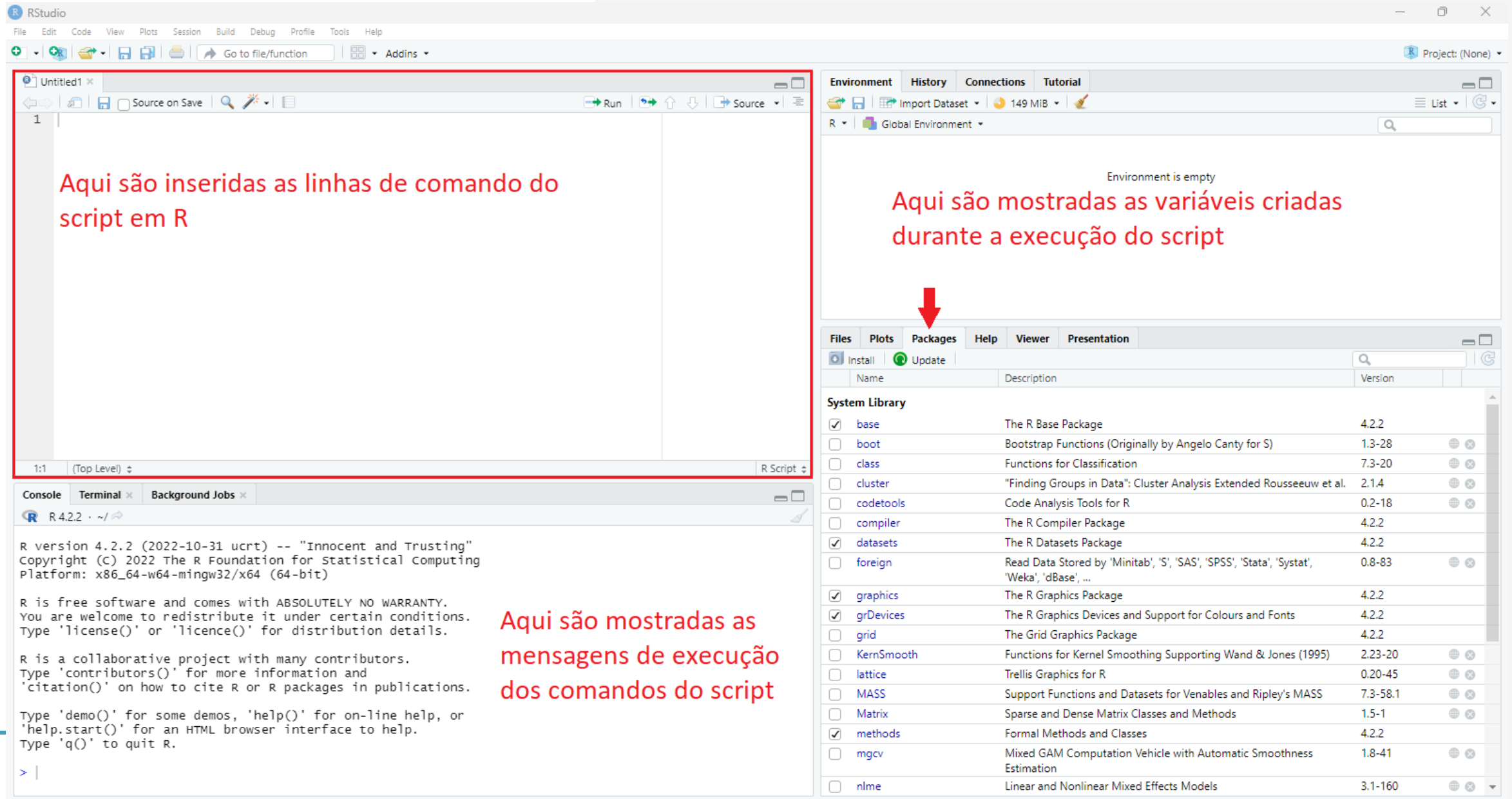
Abrir o R Studio e clicar em (+)



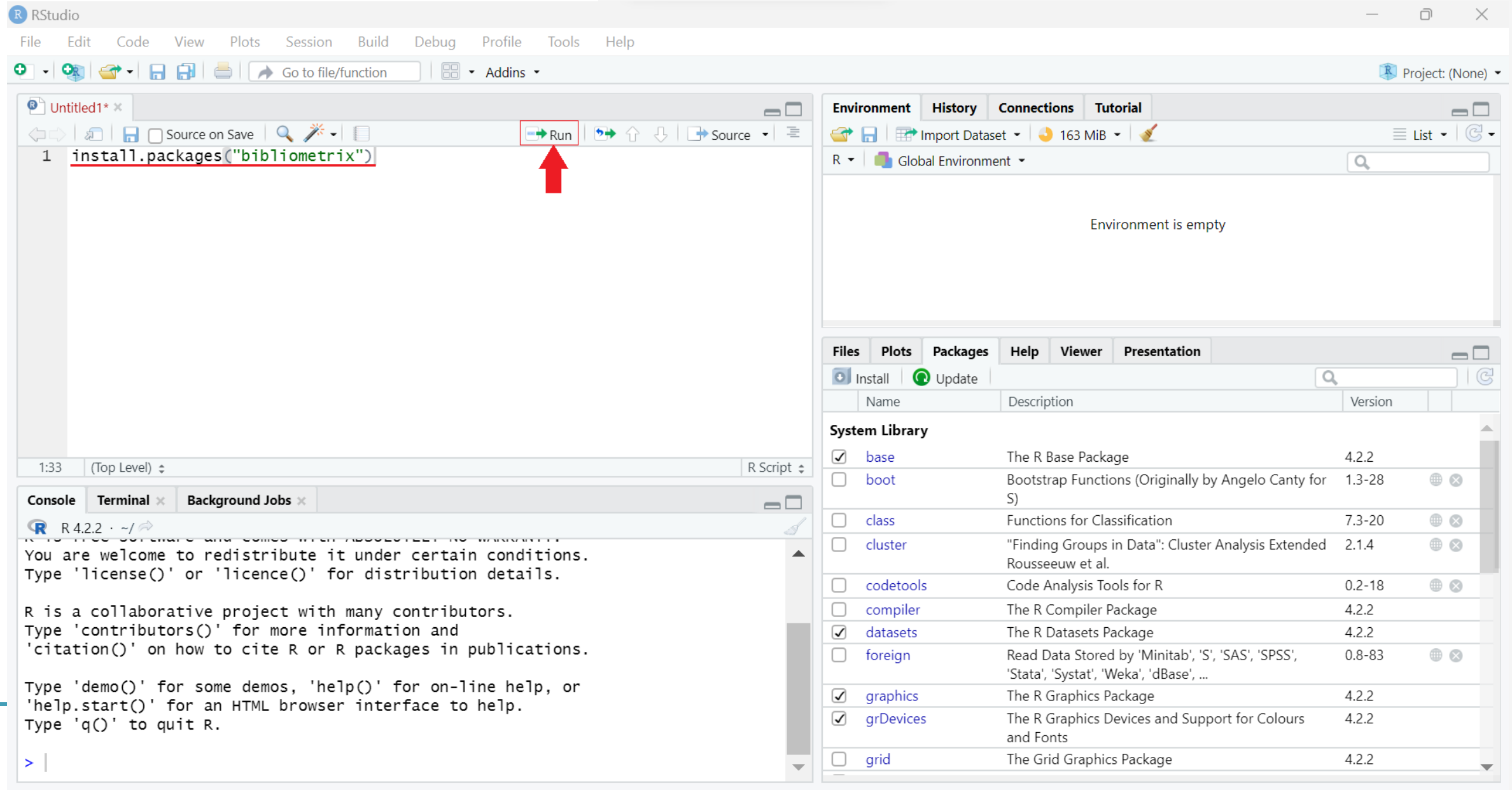
Em (+), criar um novo "R Script"



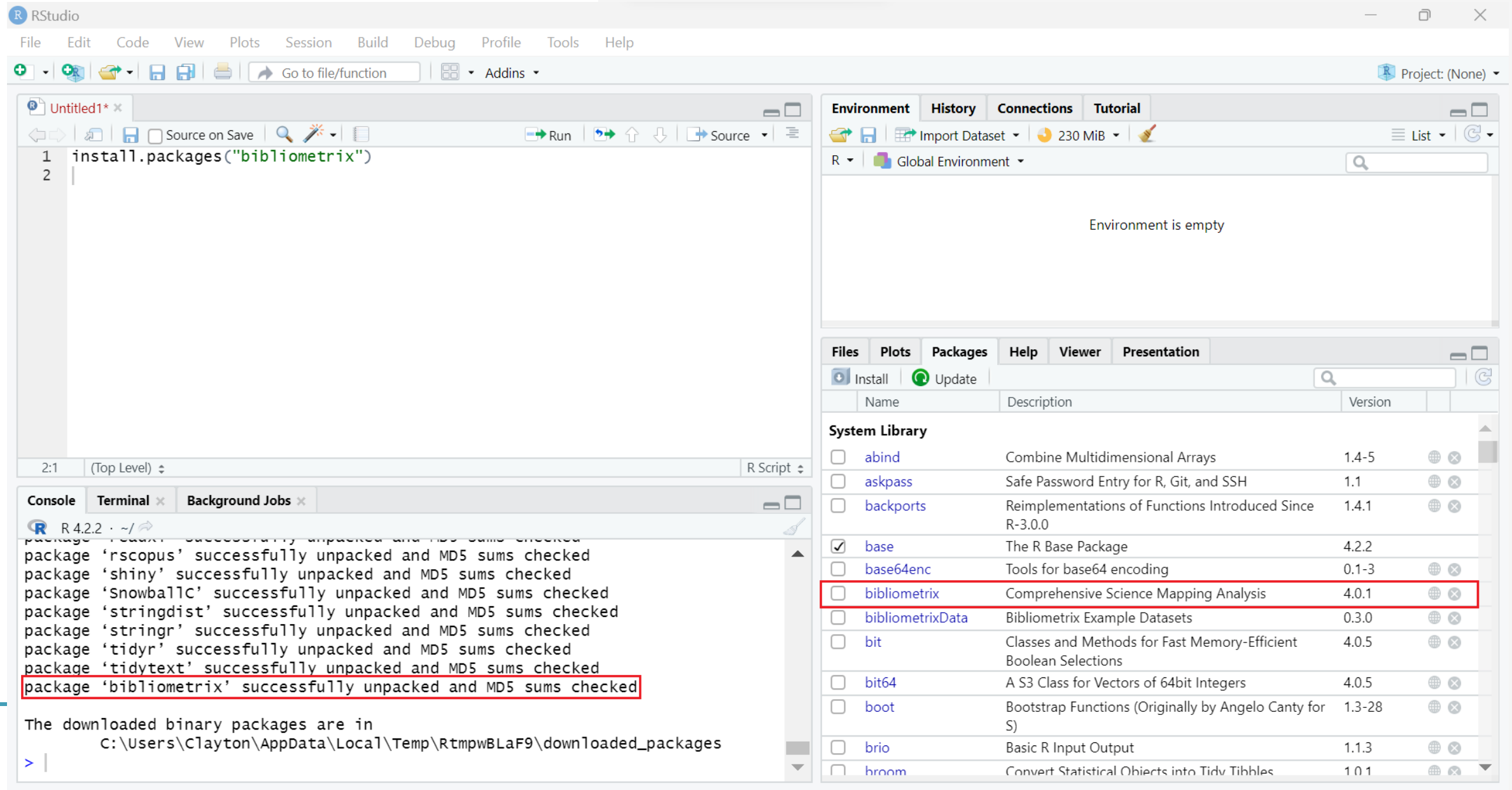
Ambiente do R Studio



Instalar o pacote Bibliometrix (apenas na 1ª vez!)



Instalar o pacote Bibliometrix (apenas na 1ª vez!)

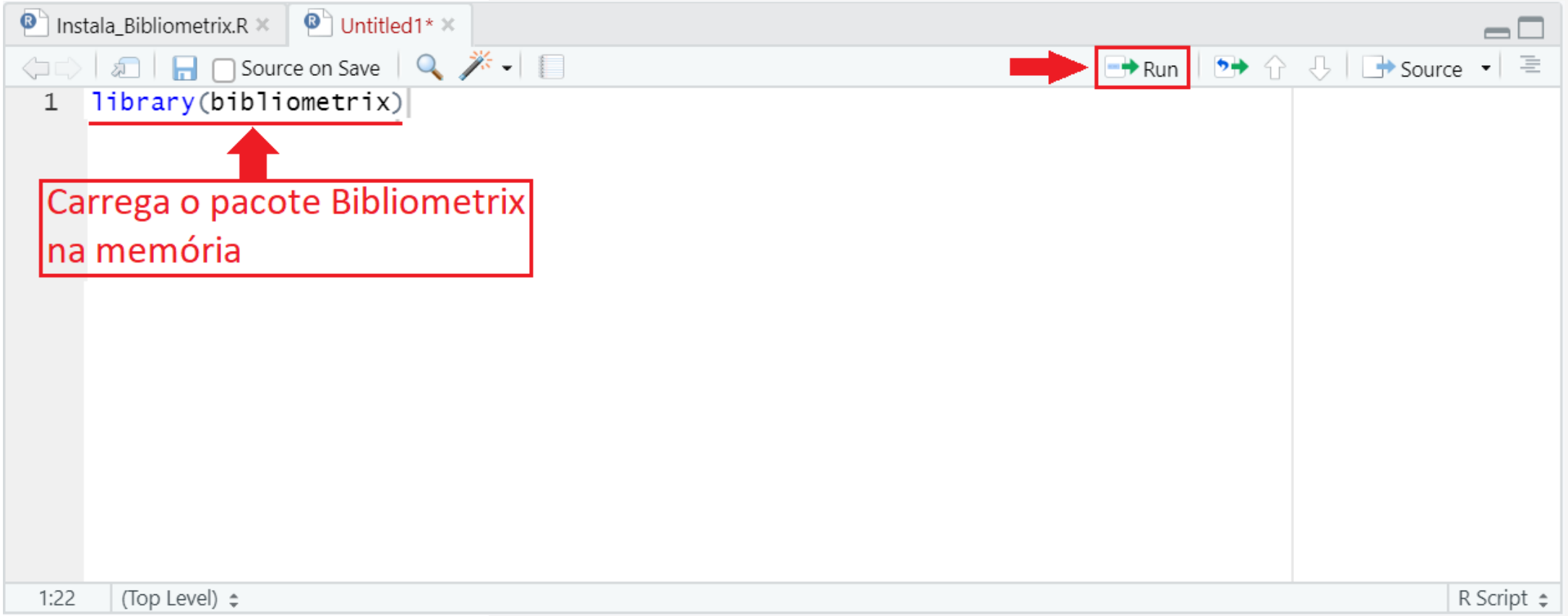


Elaboração e execução do script no R Studio

PASSO 1

Carregar o pacote “Bibliometrix” no ambiente do R Studio:

Executar o comando library(bibliometrix)



Elaboração e execução do script no R Studio

PASSO 2

a) Carregar e converter para o formato do Bibliometrix (df - *dataframe*) o conteúdo do arquivo scopus.bib:

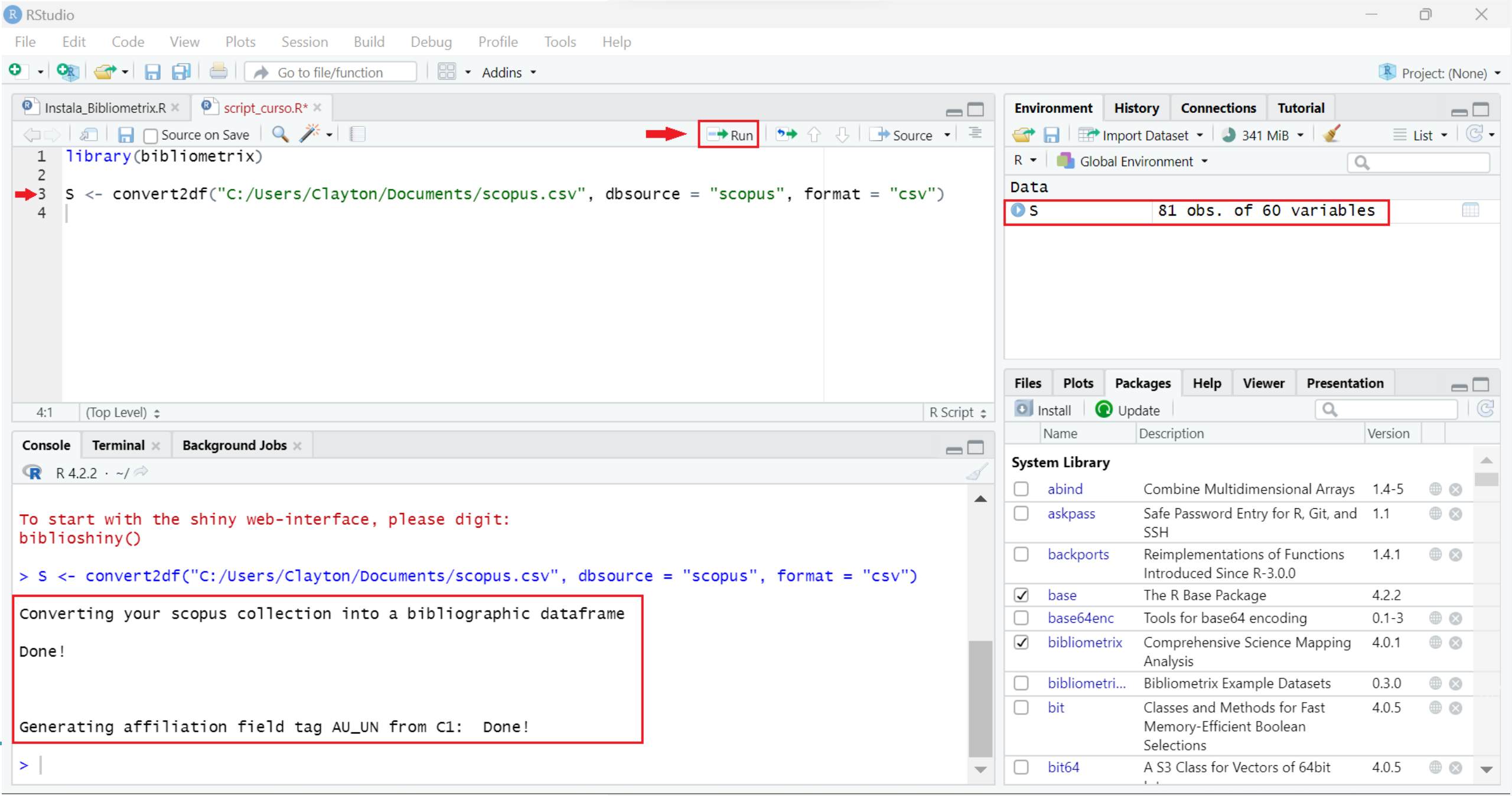
Executar o comando convert2df(*path*, dbsource, format), onde:

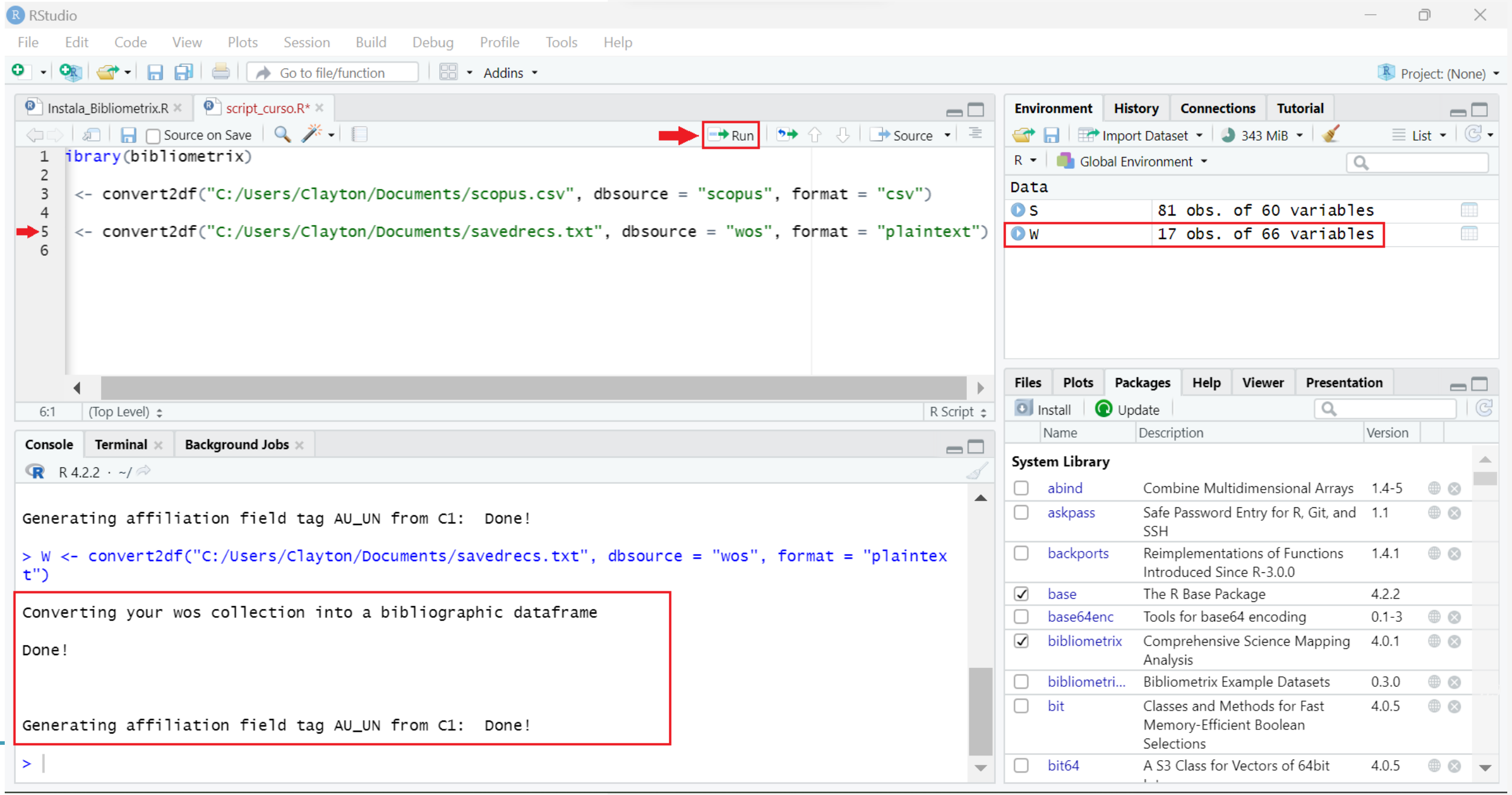
path = caminho de diretório para o arquivo de dados a ser convertido (trocar "\" por "/")

dbsource = nome da base que originou o arquivo de dados (utilizar o valor "isi" para o arquivo exportado da WoS)

format = formato de exportação do arquivo de dados (bibtex, csv, ris, ...)

b) Armazenar os dados convertidos (<-) em uma variável de ambiente (aqui "S" e "W", mas pode ser qualquer letra!)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Instala_Bibliometrix.R
script_curso.R*
Source on Save
Run
Source
1 library(bibliometrix)
2
3 S <- convert2df("C:/Users/Clayton/Documents/scopus.csv", dbsource = "scopus", format = "csv")
4
4:1 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.2.2 · ~/
To start with the shiny web-interface, please digit:
biblioshiny()
> S <- convert2df("C:/Users/Clayton/Documents/scopus.csv", dbsource = "scopus", format = "csv")
Converting your scopus collection into a bibliographic dataframe
Done!
Generating affiliation field tag AU_UN from C1: Done!
>
Environment
History
Connections
Tutorial
Import Dataset
341 MiB
List
R
Global Environment
Data
S
81 obs. of 60 variables
Files
Plots
Packages
Help
Viewer
Presentation
Install
Update
Name
Description
Version
System Library
abind
Combine Multidimensional Arrays
1.4-5
askpass
Safe Password Entry for R, Git, and SSH
1.1
backports
Reimplementations of Functions Introduced Since R-3.0.0
1.4.1
base
The R Base Package
4.2.2
base64enc
Tools for base64 encoding
0.1-3
bibliometrix
Comprehensive Science Mapping Analysis
4.0.1
bibliometri...
Bibliometrix Example Datasets
0.3.0
bit
Classes and Methods for Fast Memory-Efficient Boolean Selections
4.0.5
bit64
A S3 Class for Vectors of 64bit
4.0.5


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Instala_Bibliometrix.R
script_curso.R*
Source on Save
Run
Source
1 library(bibliometrix)
2
3 <- convert2df("C:/Users/Clayton/Documents/scopus.csv", dbsource = "scopus", format = "csv")
4
5 <- convert2df("C:/Users/Clayton/Documents/savedrecs.txt", dbsource = "wos", format = "plaintext")
6
6:1 (Top Level) R Script
Console Terminal Background Jobs
R 4.2.2 · ~/
Generating affiliation field tag AU_UN from C1: Done!
> W <- convert2df("C:/Users/Clayton/Documents/savedrecs.txt", dbsource = "wos", format = "plaintext")
Converting your wos collection into a bibliographic dataframe
Done!
Generating affiliation field tag AU_UN from C1: Done!
>
Environment History Connections Tutorial
Import Dataset 343 MiB List
R Global Environment
Data
S 81 obs. of 60 variables
W 17 obs. of 66 variables
Files Plots Packages Help Viewer Presentation
Install Update
Name Description Version
System Library
abind Combine Multidimensional Arrays 1.4-5
askpass Safe Password Entry for R, Git, and SSH 1.1
backports Reimplementations of Functions Introduced Since R-3.0.0 1.4.1
base The R Base Package 4.2.2
base64enc Tools for base64 encoding 0.1-3
bibliometrix Comprehensive Science Mapping Analysis 4.0.1
bibliometri... Bibliometrix Example Datasets 0.3.0
bit Classes and Methods for Fast Memory-Efficient Boolean Selections 4.0.5
bit64 A S3 Class for Vectors of 64bit 4.0.5

Elaboração e execução do script no R Studio

PASSO 3

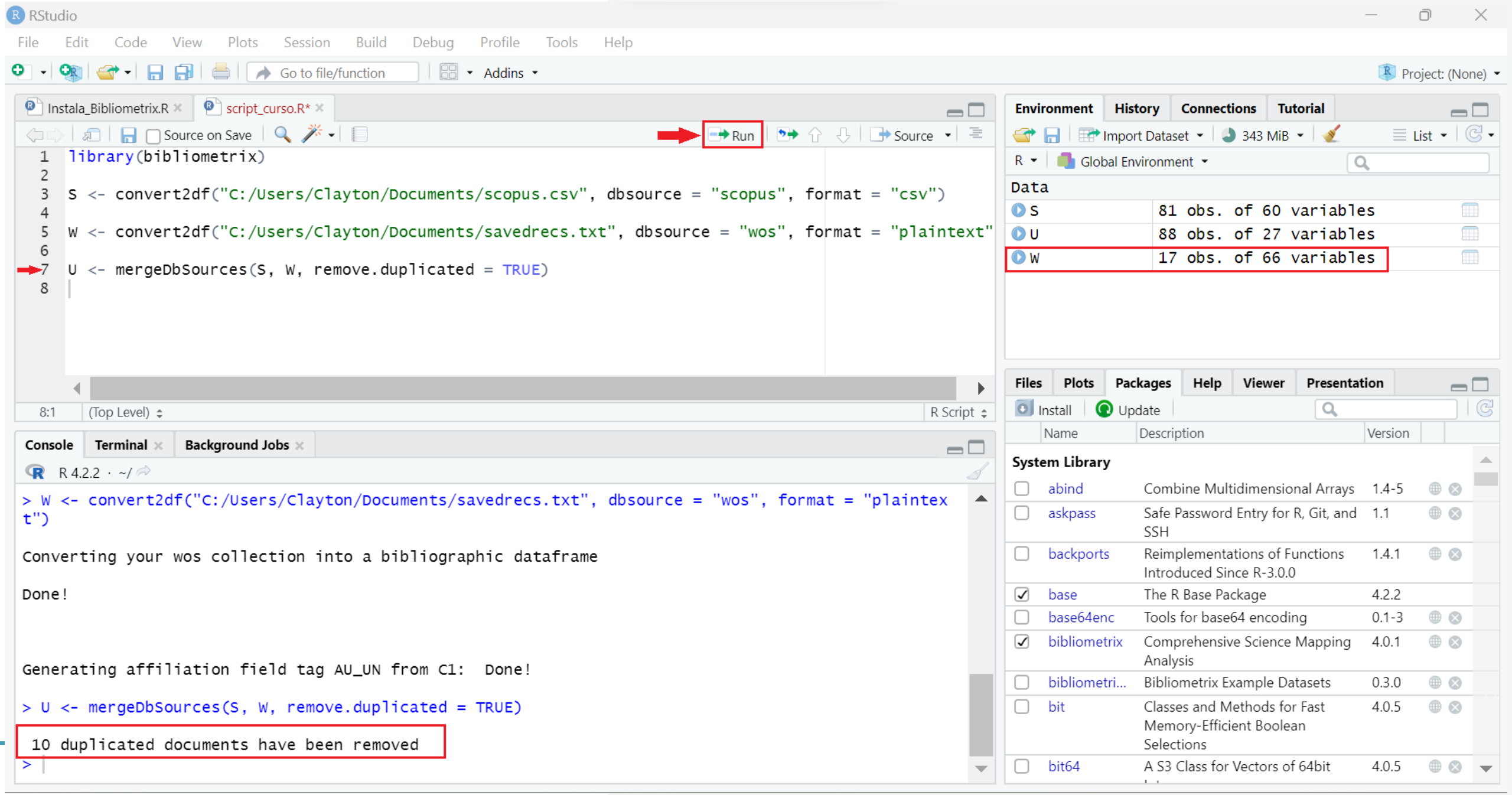
a) Unificar o conteúdo dos dois arquivos de dados convertidos e armazenados nas variáveis S e W:

Executar o comando mergeDbSources(*source1, source2,...,sourceN,* remove.duplicated), onde:

source = variáveis que contém os dados convertidos

remove.duplicated = indicar com "TRUE" ou "FALSE" se deseja, ou não, remover as referências duplicadas decorrentes da unificação dos arquivos de dados.

b) Armazenar os dados convertidos (<-) em uma variável de ambiente (aqui "U", mas pode ser qualquer letra!)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Instala_Bibliometrix.R
script_curso.R*
Source on Save
Run
Source
1 library(bibliometrix)
2
3 S <- convert2df("C:/Users/Clayton/Documents/scopus.csv", dbsource = "scopus", format = "csv")
4
5 W <- convert2df("C:/Users/Clayton/Documents/savedrecs.txt", dbsource = "wos", format = "plaintext"
6
7 U <- mergeDbSources(S, W, remove.duplicated = TRUE)
8
8:1 (Top Level) R Script
Console Terminal Background Jobs
R 4.2.2 · ~/
> W <- convert2df("C:/Users/Clayton/Documents/savedrecs.txt", dbsource = "wos", format = "plaintex
t")
Converting your wos collection into a bibliographic dataframe
Done!
Generating affiliation field tag AU_UN from C1: Done!
> U <- mergeDbSources(S, W, remove.duplicated = TRUE)
10 duplicated documents have been removed
>
Environment History Connections Tutorial
Import Dataset
343 MiB
List
R
Global Environment
Data
S 81 obs. of 60 variables
U 88 obs. of 27 variables
W 17 obs. of 66 variables
Files Plots Packages Help Viewer Presentation
Install
Update
Name Description Version
System Library
abind Combine Multidimensional Arrays 1.4-5
askpass Safe Password Entry for R, Git, and SSH 1.1
backports Reimplementations of Functions Introduced Since R-3.0.0 1.4.1
base The R Base Package 4.2.2
base64enc Tools for base64 encoding 0.1-3
bibliometrix Comprehensive Science Mapping Analysis 4.0.1
bibliometri... Bibliometrix Example Datasets 0.3.0
bit Classes and Methods for Fast Memory-Efficient Boolean Selections 4.0.5
bit64 A S3 Class for Vectors of 64bit 4.0.5

Elaboração e execução do script no R Studio

PASSO 4

a) Gravar os dados convertidos e unificados para um arquivo no formato .csv (para edição no MS-Excel)

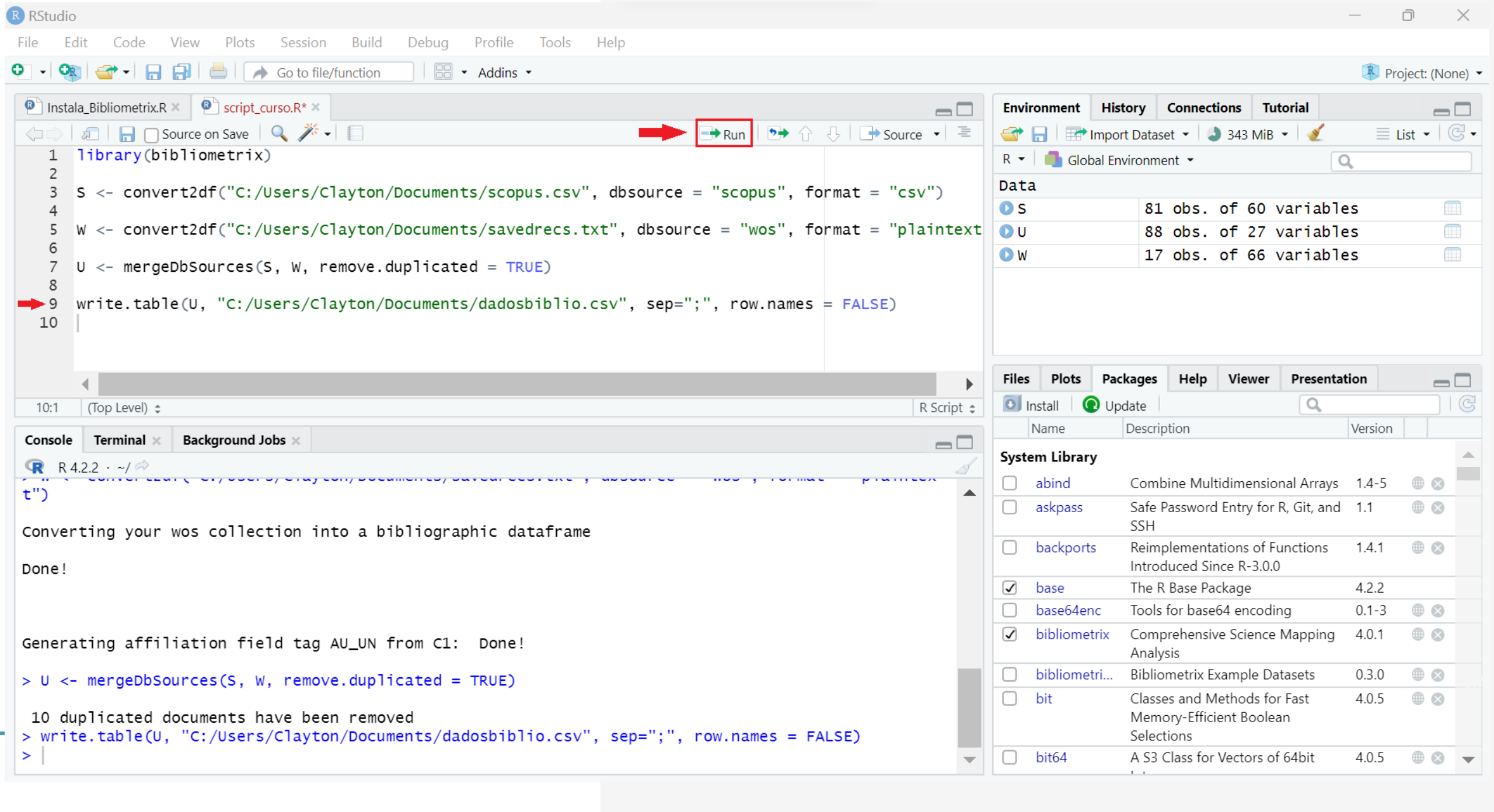
Executar o comando write.table(*source*, *path*, sep, row.names), onde:

source = variável que contém os dados unificados

path = caminho do diretório onde o arquivo .csv será gravado (trocar "\" por "/")

sep = caracter de separação das colunas de dados no arquivo a ser gerado. Normalmente é usado ";"

row.names = indicar com "TRUE" ou "FALSE" se deseja, ou não, remover os nomes de linhas no arquivo. No caso de dados da Scopus e da WoS sempre indicar "FALSE", pois essas bases não geram nomes nas linhas de dados!


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Instala_Bibliometrix.R
script_curso.R*
Source on Save
Run
Source
1 library(bibliometrix)
2
3 S <- convert2df("C:/Users/Clayton/Documents/scopus.csv", dbsource = "scopus", format = "csv")
4
5 W <- convert2df("C:/Users/Clayton/Documents/savedrecs.txt", dbsource = "wos", format = "plaintext
6
7 U <- mergeDbSources(S, W, remove.duplicated = TRUE)
8
9 write.table(U, "C:/Users/Clayton/Documents/dadosbiblio.csv", sep=";", row.names = FALSE)
10
10:1 (Top Level) R Script
Console Terminal Background Jobs
R 4.2.2 · ~/
t")
Converting your wos collection into a bibliographic dataframe
Done!
Generating affiliation field tag AU_UN from C1: Done!
> U <- mergeDbSources(S, W, remove.duplicated = TRUE)
10 duplicated documents have been removed
> write.table(U, "C:/Users/Clayton/Documents/dadosbiblio.csv", sep=";", row.names = FALSE)
>
Environment History Connections Tutorial
Import Dataset 343 MiB List
R Global Environment
Data
S 81 obs. of 60 variables
U 88 obs. of 27 variables
W 17 obs. of 66 variables
Files Plots Packages Help Viewer Presentation
Install Update
Name Description Version
System Library
abind Combine Multidimensional Arrays 1.4-5
askpass Safe Password Entry for R, Git, and SSH 1.1
backports Reimplementations of Functions Introduced Since R-3.0.0 1.4.1
base The R Base Package 4.2.2
base64enc Tools for base64 encoding 0.1-3
bibliometrix Comprehensive Science Mapping Analysis 4.0.1
bibliometri... Bibliometrix Example Datasets 0.3.0
bit Classes and Methods for Fast Memory-Efficient Boolean Selections 4.0.5
bit64 A S3 Class for Vectors of 64bit 4.0.5

Elaboração e execução do script no R Studio

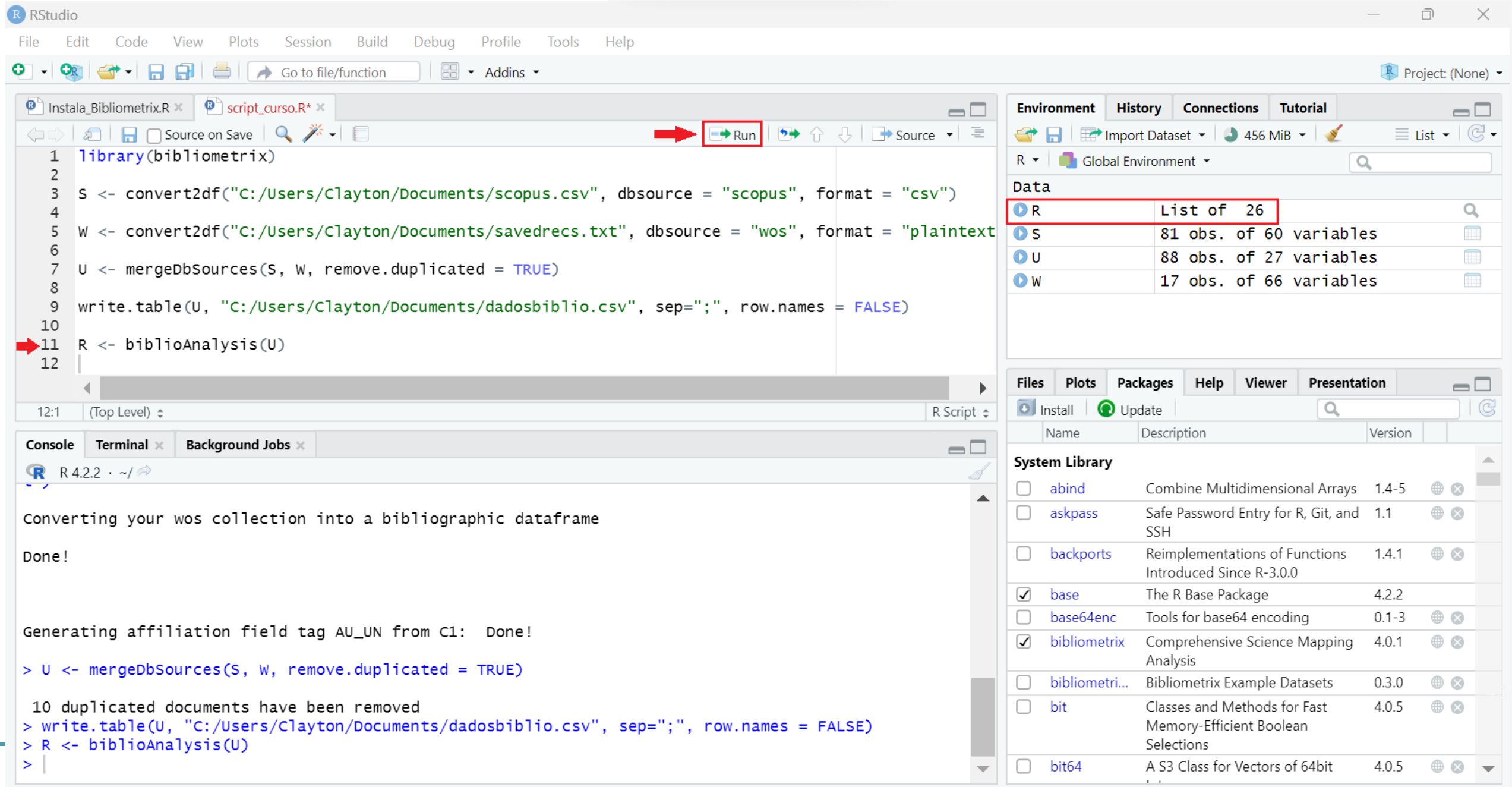
PASSO 5

a) Sumarizar os dados unificados:

Executar o comando biblIoAnalysis(*source*), onde:

source = variável que contém os dados unificados

b) Armazenar os dados convertidos (<-) em outra variável de ambiente (aqui “R”, mas pode ser qualquer letra!)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Instala_Bibliometrix.R
script_curso.R*
Source on Save
Run
Source
1 library(bibliometrix)
2
3 S <- convert2df("C:/Users/Clayton/Documents/scopus.csv", dbsource = "scopus", format = "csv")
4
5 W <- convert2df("C:/Users/Clayton/Documents/savedrecs.txt", dbsource = "wos", format = "plaintext
6
7 U <- mergeDbSources(S, W, remove.duplicated = TRUE)
8
9 write.table(U, "C:/Users/Clayton/Documents/dadosbiblio.csv", sep=";", row.names = FALSE)
10
11 R <- biblioAnalysis(U)
12
12:1 (Top Level) R Script
Console Terminal Background Jobs
R 4.2.2 · ~/
Converting your wos collection into a bibliographic dataframe
Done!
Generating affiliation field tag AU_UN from C1: Done!
> U <- mergeDbSources(S, W, remove.duplicated = TRUE)
10 duplicated documents have been removed
> write.table(U, "C:/Users/Clayton/Documents/dadosbiblio.csv", sep=";", row.names = FALSE)
> R <- biblioAnalysis(U)
>
Environment History Connections Tutorial
Import Dataset 456 MiB List
R Global Environment
Data
R List of 26
S 81 obs. of 60 variables
U 88 obs. of 27 variables
W 17 obs. of 66 variables
Files Plots Packages Help Viewer Presentation
Install Update
Name Description Version
System Library
abind Combine Multidimensional Arrays 1.4-5
askpass Safe Password Entry for R, Git, and SSH 1.1
backports Reimplementations of Functions Introduced Since R-3.0.0 1.4.1
base The R Base Package 4.2.2
base64enc Tools for base64 encoding 0.1-3
bibliometrix Comprehensive Science Mapping Analysis 4.0.1
bibliometri... Bibliometrix Example Datasets 0.3.0
bit Classes and Methods for Fast Memory-Efficient Boolean Selections 4.0.5
bit64 A S3 Class for Vectors of 64bit 4.0.5

Elaboração e execução do script no R Studio

PASSO 6

a) gerar estatísticas básicas sobre os resultados através do comando summary(object, k), onde:

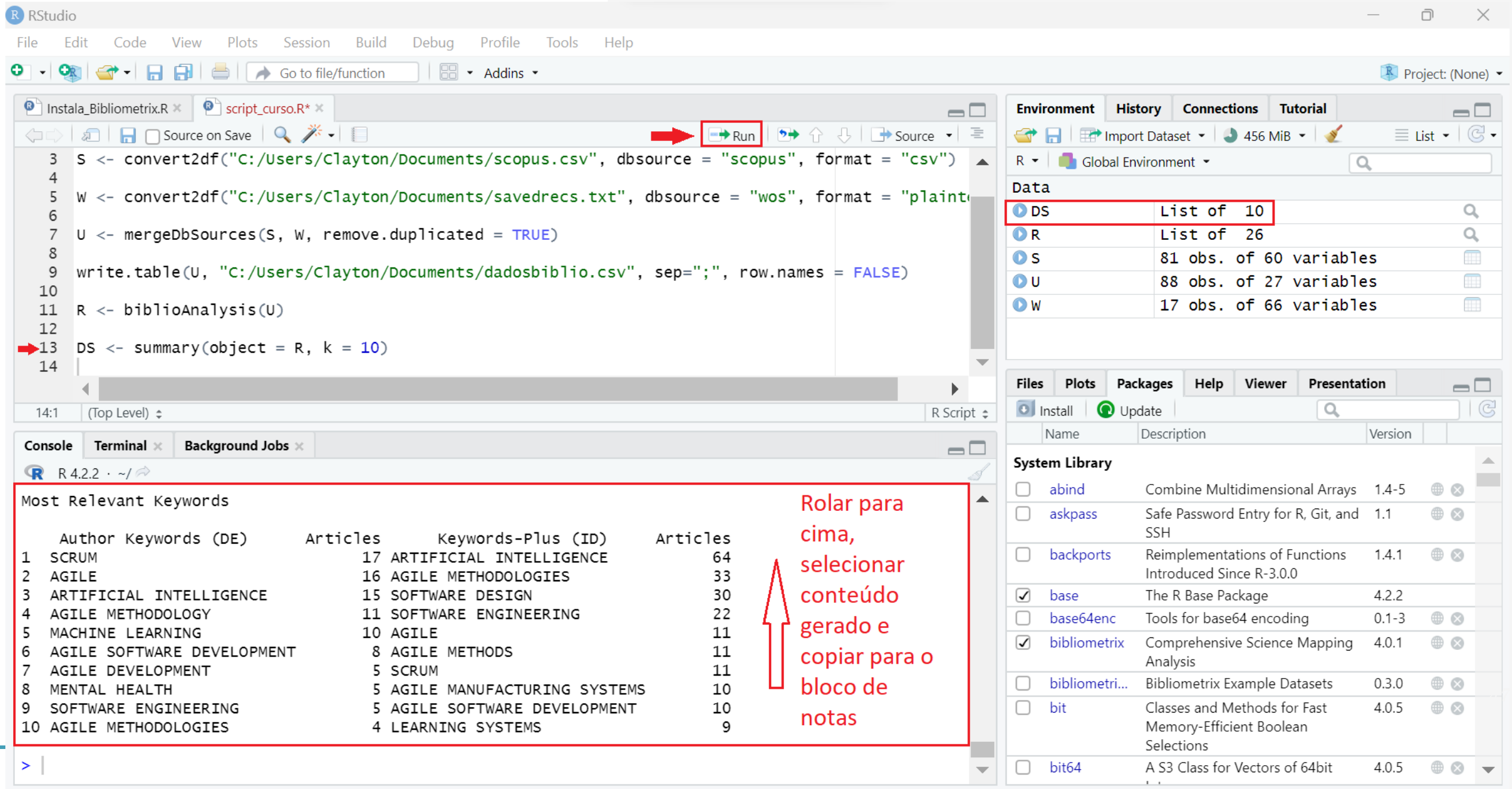
object = variável que contém a sumarização dos resultados (neste caso “R”)

k = quantidade de referências que serão exibidas (k primeiras)

b) Plotar gráficos referentes às estatísticas básicas geradas, através do comando plot(source, k), onde:

source = variável que contém a sumarização dos resultados (neste caso “R”)

k = quantidade de referências que serão exibidas (k primeiras)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Instala_Bibliometrix.R
script_curso.R*
Source on Save
Run
Source
3 S <- convert2df("C:/Users/Clayton/Documents/scopus.csv", dbsource = "scopus", format = "csv")
4
5 W <- convert2df("C:/Users/Clayton/Documents/savedrecs.txt", dbsource = "wos", format = "plaint
6
7 U <- mergeDbSources(S, W, remove.duplicated = TRUE)
8
9 write.table(U, "C:/Users/Clayton/Documents/dadosbiblio.csv", sep=";", row.names = FALSE)
10
11 R <- biblioAnalysis(U)
12
13 DS <- summary(object = R, k = 10)
14
14:1 (Top Level) R Script
Console Terminal Background Jobs
R 4.2.2 · ~/
Most Relevant Keywords
Author Keywords (DE) Articles
1 SCRUM 17
2 AGILE 16
3 ARTIFICIAL INTELLIGENCE 15
4 AGILE METHODOLOGY 11
5 MACHINE LEARNING 10
6 AGILE SOFTWARE DEVELOPMENT 8
7 AGILE DEVELOPMENT 5
8 MENTAL HEALTH 5
9 SOFTWARE ENGINEERING 5
10 AGILE METHODOLOGIES 4
Keywords-Plus (ID) Articles
1 ARTIFICIAL INTELLIGENCE 64
2 AGILE METHODOLOGIES 33
3 SOFTWARE DESIGN 30
4 SOFTWARE ENGINEERING 22
5 AGILE 11
6 AGILE METHODS 11
7 SCRUM 11
8 AGILE MANUFACTURING SYSTEMS 10
9 AGILE SOFTWARE DEVELOPMENT 10
10 LEARNING SYSTEMS 9
Rolar para cima, selecionar conteúdo gerado e copiar para o bloco de notas
>
Environment History Connections Tutorial
Import Dataset 456 MiB List
R Global Environment
Data
DS List of 10
R List of 26
S 81 obs. of 60 variables
U 88 obs. of 27 variables
W 17 obs. of 66 variables
Files Plots Packages Help Viewer Presentation
Install Update
Name Description Version
System Library
abind Combine Multidimensional Arrays 1.4-5
askpass Safe Password Entry for R, Git, and SSH 1.1
backports Reimplementations of Functions Introduced Since R-3.0.0 1.4.1
base The R Base Package 4.2.2
base64enc Tools for base64 encoding 0.1-3
bibliometrix Comprehensive Science Mapping Analysis 4.0.1
bibliometri... Bibliometrix Example Datasets 0.3.0
bit Classes and Methods for Fast Memory-Efficient Boolean Selections 4.0.5
bit64 A S3 Class for Vectors of 64bit 4.0.5

MAIN INFORMATION ABOUT DATA

Timespan	2012 : 2022
Sources (Journals, Books, etc)	70
Documents	88
Annual Growth Rate %	16.23
Document Average Age	3.8
Average citations per doc	5.568
Average citations per year per doc	1.111
References	2663

DOCUMENT TYPES

article	16
book	2
book chapter	1
conference paper	60
conference review	2
proceedings paper	5
retracted	1
review	1

DOCUMENT CONTENTS

Keywords Plus (ID)	662
Author's Keywords (DE)	293

AUTHORS

Authors	278
Author Appearances	287
Authors of single-authored docs	10

Total Citations per Country

	Country	Total Citations	Average Article Citations
1	INDIA	39	7.8
2	FINLAND	32	32.0
3	BRAZIL	29	14.5
4	UNITED KINGDOM	28	28.0
5	EGYPT	21	21.0
6	NETHERLANDS	16	8.0
7	USA	15	7.5
8	AUSTRALIA	12	4.0
9	SPAIN	9	9.0
10	DENMARK	8	8.0

Most Relevant Keywords

	Author Keywords (DE)	Articles	Keywords-Plus (ID)	Articles
1	SCRUM	17	ARTIFICIAL INTELLIGENCE	64
2	AGILE	16	AGILE METHODOLOGIES	33
3	ARTIFICIAL INTELLIGENCE	13	SOFTWARE DESIGN	30
4	AGILE METHODOLOGY	11	SOFTWARE ENGINEERING	22
5	MACHINE LEARNING	10	AGILE	11
6	AGILE SOFTWARE DEVELOPMENT	8	AGILE METHODS	11
7	AGILE DEVELOPMENT	5	SCRUM	11
8	MENTAL HEALTH	5	AGILE MANUFACTURING SYSTEMS	10
9	SOFTWARE ENGINEERING	5	AGILE SOFTWARE DEVELOPMENT	10
10	AGILE METHODOLOGIES	4	LEARNING SYSTEMS	9

AUTHORS COLLABORATION

Single-authored docs	12
Documents per Author	0.317
Co-Authors per Doc	3.26
International co-authorships %	0

Annual Scientific Production

Year	Articles
2012	2
2013	3
2014	8
2015	2
2016	8
2017	4
2018	15
2019	17
2020	10
2021	10
2022	9

Annual Percentage Growth Rate 16.23081

Most Productive Authors

	Authors	Articles	Authors	Articles Fractionalized
1	ANDRADE-ARENAS L	2	MEYER M	2
2	BAGRIYANIK S	2	NA N	2
3	FUERTES W	2	CASEAU Y	1
4	KUMAR D	2	CHOI J	1
5	MEYER M	2	FIRETEANU V	1
6	NA N	2	IQBAL H	1
7	SARIKAYA E	2	JANAKOVA M	1
8	VILLACÍS C	2	LYNGGAARD P	1
9	ZAMBRANO M	2	MARCHEGIANI L	1
10	ABBAS M	1	SCHLEIER-SMITH J	1

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

Instala_Bibliometrix.R | script_curso.R*

Source on Save | Run | Source

```
5  W <- convert2df("C:/Users/Clayton/Documents/savedrecs.txt", dbsource = "wos", format = "plaint
6
7  U <- mergeDbSources(S, W, remove.duplicated = TRUE)
8
9  write.table(U, "C:/Users/Clayton/Documents/dadosbiblio.csv", sep=";", row.names = FALSE)
10
11 R <- biblioAnalysis(U)
12
13 DS <- summary(object = R, k = 10)
14
15 plot(R, k = 10)
16
```

16:1 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.2.2 · ~/

1	SCRUM	17	ARTIFICIAL INTELLIGENCE	64
2	AGILE	16	AGILE METHODOLOGIES	33
3	ARTIFICIAL INTELLIGENCE	15	SOFTWARE DESIGN	30
4	AGILE METHODOLOGY	11	SOFTWARE ENGINEERING	22
5	MACHINE LEARNING	10	AGILE	11
6	AGILE SOFTWARE DEVELOPMENT	8	AGILE METHODS	11
7	AGILE DEVELOPMENT	5	SCRUM	11
8	MENTAL HEALTH	5	AGILE MANUFACTURING SYSTEMS	10
9	SOFTWARE ENGINEERING	5	AGILE SOFTWARE DEVELOPMENT	10
10	AGILE METHODOLOGIES	4	LEARNING SYSTEMS	9

```
> plot(R, k = 10)
Warning message:
Removed 1 rows containing non-finite values (`stat_align()`).
>
```

Environment | History | Connections | Tutorial

Import Dataset | 608 MiB | List

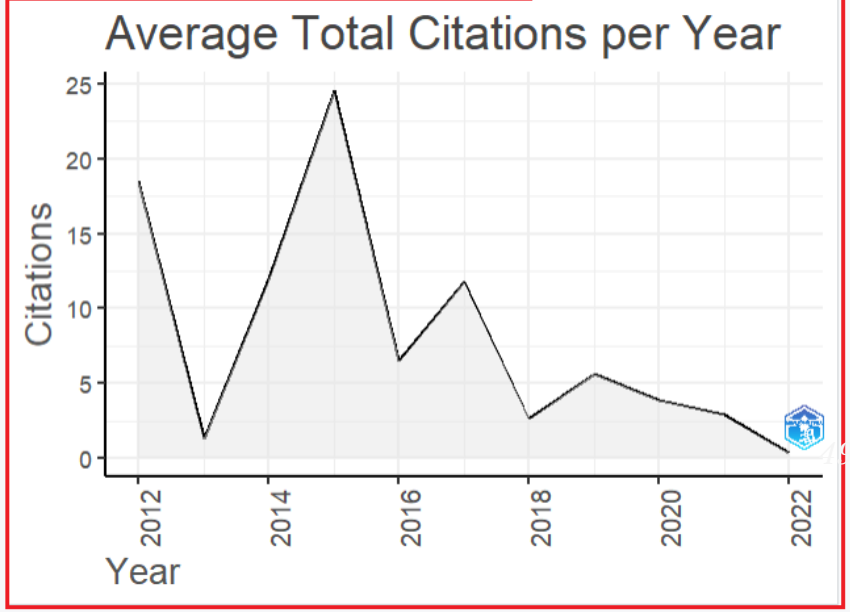
R | Global Environment

Data

DS	List of 10
R	List of 26
S	81 obs. of 60 variables
U	88 obs. of 27 variables
W	17 obs. of 66 variables

Files | Plots | Packages | Help | Viewer | Presentation

Zoom | Export



Análises bibliométricas com biblioshiny

- A *biblioshiny* é uma ferramenta, incluída no pacote *Bibliometrix*, que possibilita a geração e visualização de diversas análises bibliométricas em arquivos de dados bibliográficos.
- É compatível com os browsers web ***Mozilla Firefox*** e ***Google Chrome***.
- Pode receber como entrada arquivos de dados bibliográficos nos formatos **BibTeX** ou **XLSX**. Por isso, o arquivo .csv contendo os dados unificados no script R (comando *write.table*), deve ser aberto no MS-Excel e salvo como planilha XLSX antes de ser usado na *biblioshiny*!
- Para executar a ferramenta, no *R Studio*, execute o comando biblioshiny().

Análises bibliométricas com biblioshiny

- Devido à carga horária do curso, serão apresentadas as análises geradas pela ferramenta sobre os dados utilizados como exemplo, porém não serão aprofundados os conceitos estatísticos e bibliométricos empregados em tais análises.
- Para maiores detalhes sobre as análises bibliométricas efetuadas pela ferramenta, consultar as referências a seguir:
  - Aria, M. & Cuccurullo, C. (2017). bibliometrix: An R-tool for comprehensive science mapping analysis, Journal of Informetrics, 11(4), pp 959-975, Elsevier, DOI: 10.1016/j.joi.2017.08.007.
  - A tutorial about the main functions implemented in bibliometrix: https://www.bibliometrix.org/vignettes/Introduction_to_bibliometrix.html
  - A tutorial about "how to import data from the main databases": https://www.bibliometrix.org/vignettes/Data-Importing-and-Converting.html
  - A tutorial about "how to use Biblioshiny": https://bibliometrix.org/biblioshiny/assets/player/KeynoteDHTMLPlayer.html

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

Instala_Bibliometrix.R | script_curso.R*

Source on Save | Run | Source

```
 7  U <- mergeDbSources(S, W, remove.duplicated = TRUE)
 8
 9  write.table(U, "C:/Users/Clayton/Documents/dadosbiblio.csv", sep=";", row.names = FALSE)
10
11  R <- biblioAnalysis(U)
12
13  DS <- summary(object = R, k = 10)
14
15  plot(R, k = 10)
16
17  biblioshiny()
18
```

18:1 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.2.2 · ~/

```
4  AGILE METHODOLOGY                11 SOFTWARE ENGINEERING              22
5  MACHINE LEARNING                 10 AGILE                             11
6  AGILE SOFTWARE DEVELOPMENT        8 AGILE METHODS                     11
7  AGILE DEVELOPMENT                 5 SCRUM                             11
8  MENTAL HEALTH                     5 AGILE MANUFACTURING SYSTEMS       10
9  SOFTWARE ENGINEERING              5 AGILE SOFTWARE DEVELOPMENT        10
10 AGILE METHODOLOGIES               4 LEARNING SYSTEMS                   9

> plot(R, k = 10)
Warning message:
Removed 1 rows containing non-finite values (`stat_align()`).
> biblioshiny()
Carregando pacotes exigidos: shiny

Listening on http://127.0.0.1:4443
```

Environment | History | Connections | Tutorial

Import Dataset | 645 MiB | List

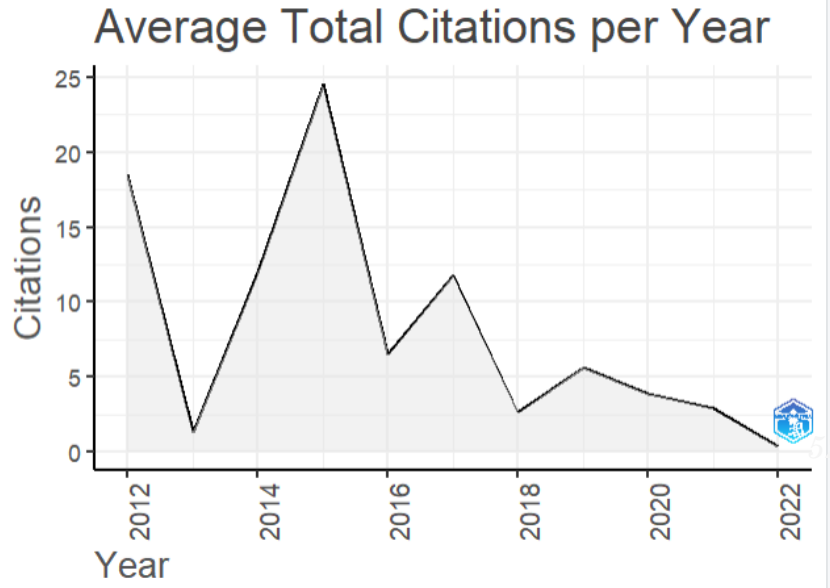
R | Global Environment

Data

DS	List of 10
R	List of 26
S	81 obs. of 60 variables
U	88 obs. of 27 variables
W	17 obs. of 66 variables

Files | Plots | Packages | Help | Viewer | Presentation

Zoom | Export



127.0.0.1:7899/

127.0.0.1:7899

bibliometrix

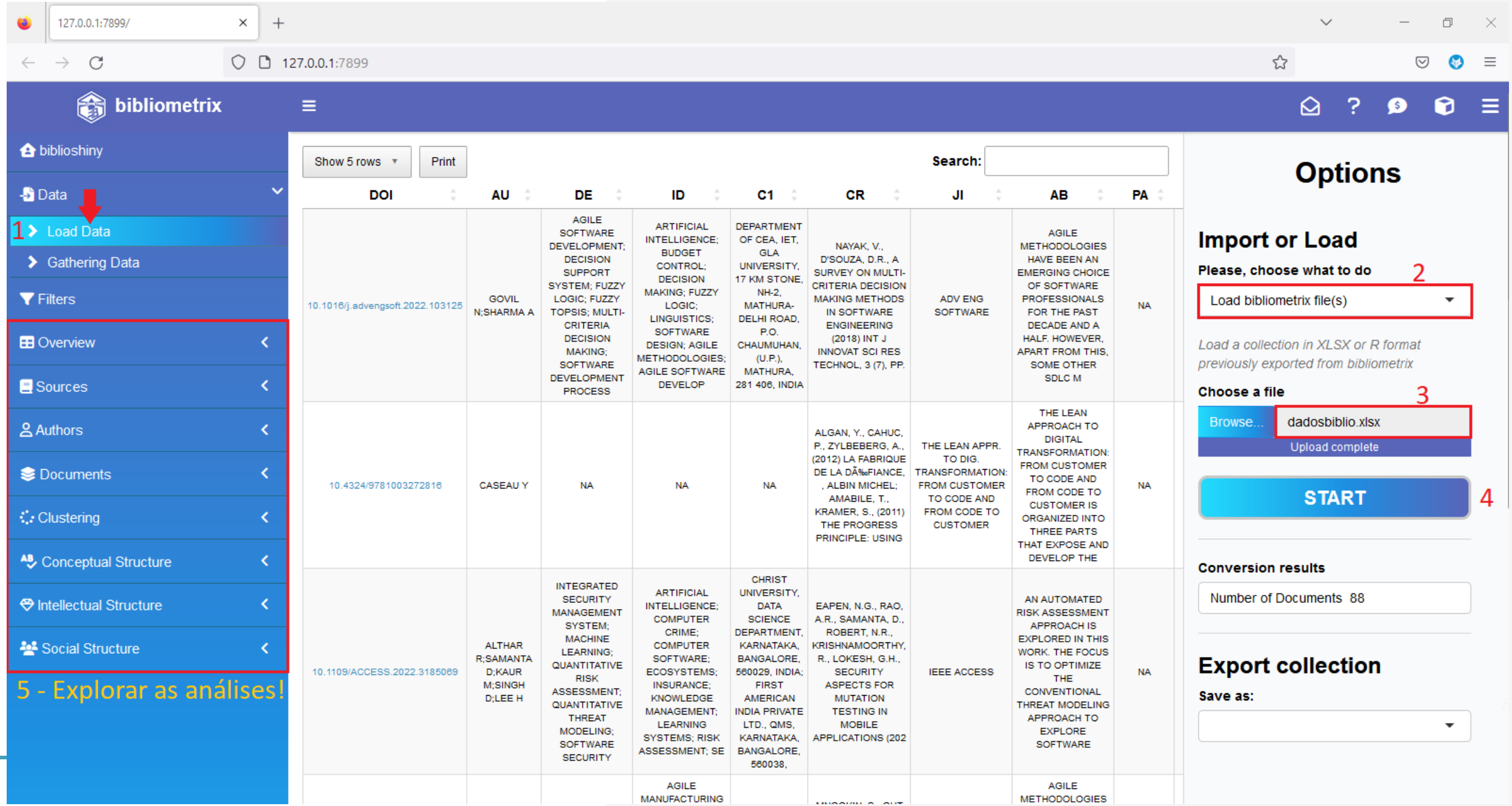
biblioshiny

Data

biblioshiny: the shiny app for bibliometrix

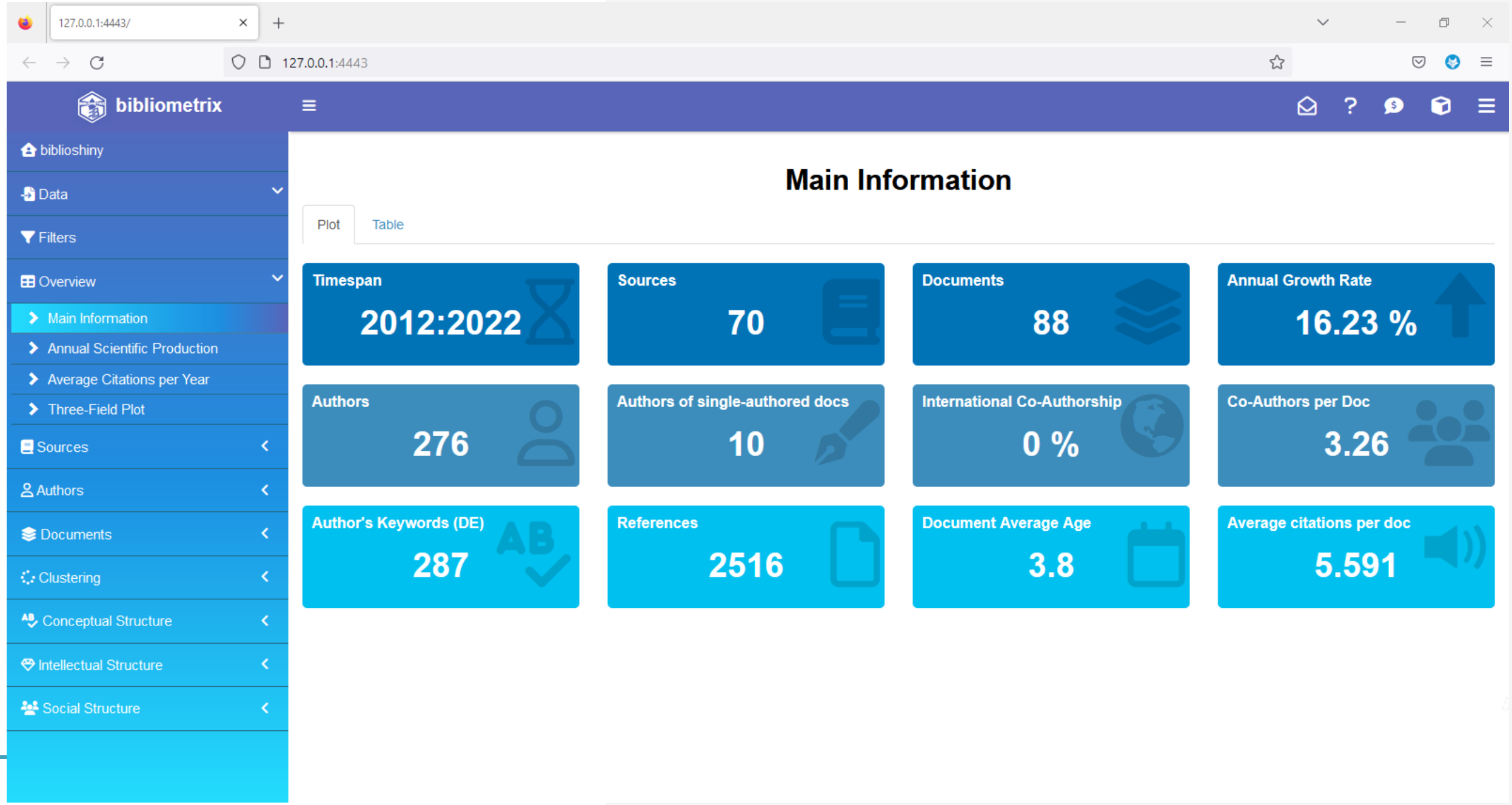
Aria, M., & Cuccurullo, C. (2017). **bibliometrix: An R-tool for comprehensive science mapping analysis.** *Journal of Informetrics* , 11(4), 959-975.

For an introduction and live examples, visit the *bibliometrix website*


127.0.0.1:7899/
127.0.0.1:7899
bibliometrix
biblioshiny
Data
1
Load Data
Gathering Data
Filters
Overview
Sources
Authors
Documents
Clustering
Conceptual Structure
Intellectual Structure
Social Structure
5 - Explorar as análises!
Show 5 rows
Print
Search:
DOI
AU
DE
ID
C1
CR
JI
AB
PA
10.1016/j.advengsoft.2022.103125
GOVIL N;SHARMA A
AGILE SOFTWARE DEVELOPMENT; DECISION SUPPORT SYSTEM; FUZZY LOGIC; FUZZY TOPSIS; MULTI-CRITERIA DECISION MAKING; SOFTWARE DEVELOPMENT PROCESS
ARTIFICIAL INTELLIGENCE; BUDGET CONTROL; DECISION MAKING; FUZZY LOGIC; LINGUISTICS; SOFTWARE DESIGN; AGILE METHODOLOGIES; AGILE SOFTWARE DEVELOP
DEPARTMENT OF CEA, IET, GLA UNIVERSITY, 17 KM STONE, NH-2, MATHURA-DELHI ROAD, P.O. CHAUMUHAN, (U.P.), MATHURA, 281 406, INDIA
NAYAK, V., D'SOUZA, D.R., A SURVEY ON MULTI-CRITERIA DECISION MAKING METHODS IN SOFTWARE ENGINEERING (2018) INT J INNOVAT SCI RES TECHNOL, 3 (7), PP.
ADV ENG SOFTWARE
AGILE METHODOLOGIES HAVE BEEN AN EMERGING CHOICE OF SOFTWARE PROFESSIONALS FOR THE PAST DECADE AND A HALF. HOWEVER, APART FROM THIS, SOME OTHER SDLC M
NA
10.4324/9781003272816
CASEAU Y
NA
NA
NA
ALGAN, Y., CAHUC, P., ZYLBEBERG, A., (2012) LA FABRIQUE DE LA DÂ‰FIANCE, , ALBIN MICHEL; AMABILE, T., KRAMER, S., (2011) THE PROGRESS PRINCIPLE: USING
THE LEAN APPR. TO DIG. TRANSFORMATION: FROM CUSTOMER TO CODE AND FROM CODE TO CUSTOMER
THE LEAN APPROACH TO DIGITAL TRANSFORMATION: FROM CUSTOMER TO CODE AND FROM CODE TO CUSTOMER IS ORGANIZED INTO THREE PARTS THAT EXPOSE AND DEVELOP THE
NA
10.1109/ACCESS.2022.3185069
ALTHAR R;SAMANTA D;KAUR M;SINGH D;LEE H
INTEGRATED SECURITY MANAGEMENT SYSTEM; MACHINE LEARNING; QUANTITATIVE RISK ASSESSMENT; QUANTITATIVE THREAT MODELING; SOFTWARE SECURITY
ARTIFICIAL INTELLIGENCE; COMPUTER CRIME; COMPUTER SOFTWARE; ECOSYSTEMS; INSURANCE; KNOWLEDGE MANAGEMENT; LEARNING SYSTEMS; RISK ASSESSMENT; SE
CHRIST UNIVERSITY, DATA SCIENCE DEPARTMENT, KARNATAKA, BANGALORE, 560029, INDIA; FIRST AMERICAN INDIA PRIVATE LTD., QMS, KARNATAKA, BANGALORE, 560038,
EAPEN, N.G., RAO, A.R., SAMANTA, D., ROBERT, N.R., KRISHNAMOORTHY, R., LOKESH, G.H., SECURITY ASPECTS FOR MUTATION TESTING IN MOBILE APPLICATIONS (202
IEEE ACCESS
AN AUTOMATED RISK ASSESSMENT APPROACH IS EXPLORED IN THIS WORK. THE FOCUS IS TO OPTIMIZE THE CONVENTIONAL THREAT MODELING APPROACH TO EXPLORE SOFTWARE
NA
AGILE MANUFACTURING
AGILE METHODOLOGIES
Options
Import or Load
Please, choose what to do
2
Load bibliometrix file(s)
Load a collection in XLSX or R format previously exported from bibliometrix
Choose a file
3
Browse...
dadosbiblio.xlsx
Upload complete
START
4
Conversion results
Number of Documents 88
Export collection
Save as:

Códigos de campos

AB	Abstract	ID	Indexer Keywords
AF	Author Full Name	IS	Issue
AR	Article Number	J9	29-Character Source Abbreviation
AU	Authors	JI	ISO Source Abbreviation
BA	Book Authors	LA	Language
BE	Editors	MA	Meeting Abstract
BF	Book Authors Full Name	NR	Cited Reference Count
BN	International Standard Book Number (ISBN)	OA	Open Access Indicator
BP	Beginning Page	OI	ORCID Identifier (Open Researcher and Contributor ID)
BS	Book Series Subtitle	P2	Chapter Count (Book Citation Index)
C1	Author Address	PA	Publisher Address
CA	Group Authors	PD	Publication Date
CL	Conference Location	PG	Page Count
CR	Cited References	PI	Publisher City
CT	Conference Title	PM	PubMed ID
CY	Conference Date	PN	Part Number
D2	Book Digital Object Identifier (DOI)	PT	Publication Type (J=Journal; B=Book; S=Series; P=Patent)
DA	Date this report was generated.	PU	Publisher
DE	Author Keywords	PY	Year Published
DI	Digital Object Identifier (DOI)	RI	ResearcherID Number
DT	Document Type	RP	Reprint Address
EA	Early access date	SC	Research Areas
EF	End of File	SE	Book Series Title
EI	Electronic International Standard Serial Number (eISSN)	SI	Special Issue
EM	E-mail Address	SN	International Standard Serial Number (ISSN)
EP	Ending Page	SO	Publication Name
ER	End of Record	SP	Conference Sponsors
EY	Early access year	SU	Supplement
FN	File Name	TC	Web of Science Core Collection Times Cited Count
FU	Funding Agency and Grant Number	TI	Document Title
FX	Funding Text	U1	Usage Count (Last 180 Days)
GA	Document Delivery Number	U2	Usage Count (Since 2013)
GP	Book Group Authors	UT	Accession Number
HC	ESI Highly Cited Paper. Note that this field is valued only for ESI subscribers.	VL	Volume
HO	Conference Host	VR	Version Number
HP	ESI Hot Paper. Note that this field is valued only for ESI subscribers.	WC	Web of Science Categories
		Z9	Total Times Cited Count (Web of Science Core Collection, BIOSIS Citation Index, Chinese Science Citation Database, Data Citation Index, Russian Science Citation Index, SciELO Citation Index)


127.0.0.1:4443/
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bibliometrix
biblioshiny
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Three-Field Plot
Sources
Authors
Documents
Clustering
Conceptual Structure
Intellectual Structure
Social Structure
Main Information
Plot
Table
Timespan
2012:2022
Sources
70
Documents
88
Annual Growth Rate
16.23 %
Authors
276
Authors of single-authored docs
10
International Co-Authorship
0 %
Co-Authors per Doc
3.26
Author's Keywords (DE)
287
References
2516
Document Average Age
3.8
Average citations per doc
5.591


127.0.0.1:4443/
127.0.0.1:4443
bibliometrix
biblioshiny
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Filters
Overview
Sources
Authors
Documents
Documents
Most Global Cited Documents
Most Local Cited Documents
Cited References
Most Local Cited References
Reference Spectroscopy
Words
Most Frequent Words
WordCloud
TreeMap
Word Dynamics
Trend Topics
Clustering
WordCloud
Run
Plot
Table
human resource management
decision support systems
international trade
machine-learning
requirements engineering
learning systems
iterative methods
software requirements
agile manufacturing systems
sentiment analysis
integration testing
continuous integrations
computers
digital storage
scrum agile
software company
software development process
teaching
decision making
software design
software industry
n.a
software testing
digital transformation
automation
students
education
agile methodologies
computer software
data analytics
internet of things
life cycle
e-learning
engineering education
software engineering
surveys
computer science
agile methods
agile development
big data
teaching and learning
genetic counseling
agile software development
software project
project management
machine learning
application programs
software development methodologies
Options
Field
Keywords Plus
Number of words
50
Text Editing
Parameters

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bibliometrix

- biblioshiny
- Data
- Filters
- Overview
- Sources
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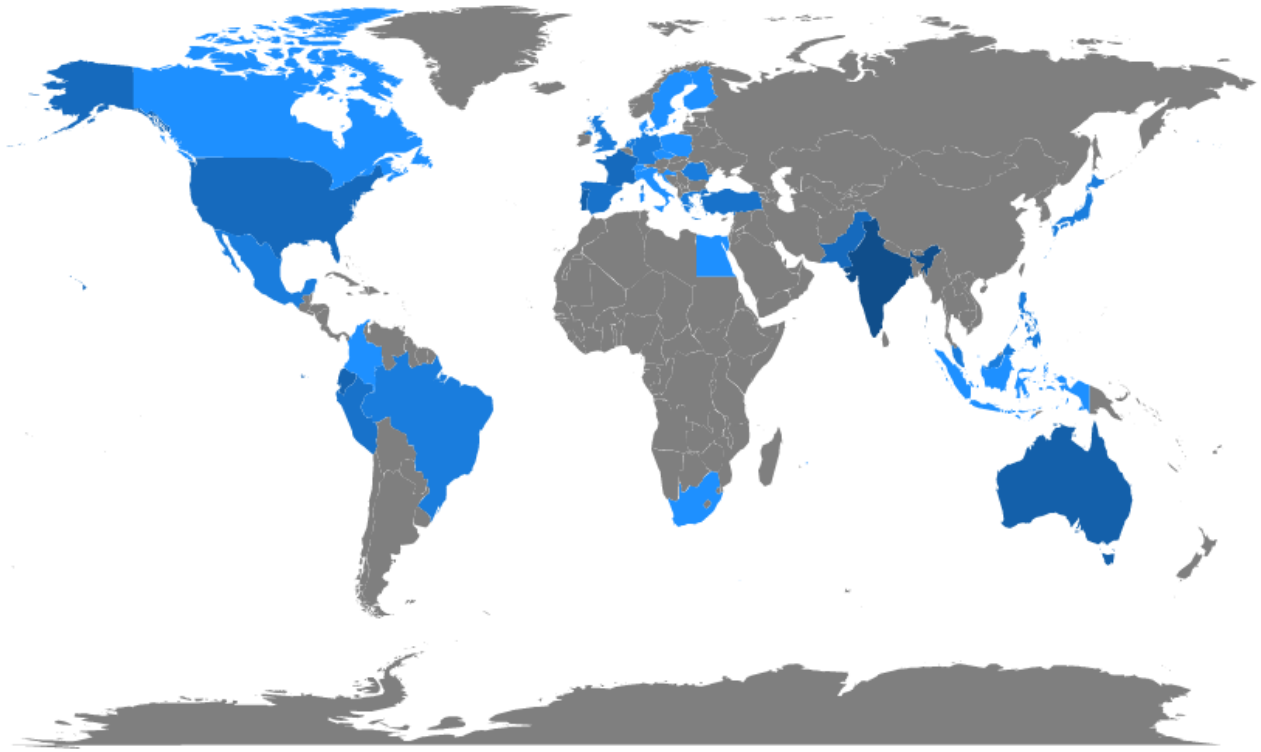
Authors
- Most Relevant Authors
- Most Local Cited Authors
- Authors' Production over Time
- Lotka's Law
- Author Impact

Affiliations
- Most Relevant Affiliations
- Affiliations' Production over Time

Countries
- Corresponding Author's Country
- Country Scientific Production
- Countries' Production over Time
- Most Cited Countries

Country Scientific Production

Plot | Table



Options

Export plot

dpi value

Final do curso

- Dúvidas sobre o conteúdo deste curso e de resolução do exercício proposto, entrar em contato pelo e-mail:

martins.claytonp@gmail.com

- Mais sobre o *Bibliometrix* e *biblioshiny*:

https://www.bibliometrix.org



Bibliometria na Prática com *R* e *Bibliometrix*

Coleta de dados nas bases *Scopus* e *Web of Science*

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FFC/UNESP

dezembro/2022